JÖNKÖPING UNIVERSITY
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Barriers to Environmental Sustainability in EU Steel and Metal Manufacturing SMEs

A qualitative study on how barriers constrain environmental sustainability in relation to Sensing, Seizing, and Transforming capabilities within steel and metal manufacturing SMEs in the EU

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Abstract

Background: Environmental sustainability has become important for manufacturing SMEs, particularly for those that operate in a resource-intensive sector such as the steel and metal manufacturing industry. SMEs contribute significantly to economic value creation. They are also under increasing pressure from customers, regulatory bodies, certification systems and actors in the supply-chain to reduce their environmental impact. However, these firms often have limited financial resources, weak data systems while facing organizational capacity constraints.

Purpose: The purpose of this thesis is to examine how barriers appear to constrain the environmental sustainability of steel and metal manufacturing SMEs embedded in the EU, using the Dynamic Capabilities theory as an exploratory lens to explore key mechanisms.

Method: The study applies a qualitative research design based on semi-structured interviews with four Hungarian steel and metal manufacturing SMEs embedded in EU-wide supply chains. The empirical material was analyzed on a thematic basis through a codebook, developed based on internal barriers, external barriers, sustainability reporting, the Dynamic Capabilities of sensing, seizing and transforming, and emerging themes.

Conclusion: Results indicate that environmental sustainability in the studied SMEs is restricted by a system of interrelated barriers. Sensing is fragmented, seizing is selective, transformation is partial, the movement between capabilities is constrained, and sustainability reporting may act as an interface bottleneck. Overall, the findings suggest that sustainability barriers may constrain the development of sustainability-related Dynamic Capabilities that support environmental sustainability transition.

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1. Introduction

This section of the thesis contains the background of our research introducing the importance of SMEs from both sustainability and economic perspectives. It is followed by an outlined problem discussion about the challenges of environmental sustainability in manufacturing SMEs. Thirdly, it introduces the purpose, and the intended research question while indicating the delimitations of the thesis

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1.1 Background

Environmental sustainability and emissions reduction have become increasingly urgent and prominent aspects in global discourse throughout the past years. International bodies including the United Nations and the European Union are initiating immediate action across all industries and sectors with the aim of addressing climate crises (Dragomir et al., 2022). These initiatives are more important than ever as global warming has reached 1.2 °C (Sohns et al., 2023) and is expected to exceed the Paris Agreement target of keeping global warming below 1.5 °C compared to pre-industrial levels (United Nations, 2015). Therefore, it is essential to act towards sustainable business operations across all sectors and all levels of the economy in which small and medium-sized enterprises (SMEs) play an important role.

SMEs are important as they constitute over 95% of the companies worldwide while also accounting for more than 60% of private sector employment (Alayón et al., 2022). Moreover, when we consider companies collectively across Europe, the results indicate that SMEs are not only the most common form of businesses, but they also add more than 56% of the total value to the European Union's economy (Sohns et al., 2023). However, adding value is not the only aspect that makes SMEs essential. Their environmental footprint adds up to 64% of all industrial pollution in the European Union and 70% of pollution worldwide (Alayón et al., 2022; Neri et al., 2021; Yadav & Gahlot, 2022) while also operating with high energy consumption, as they account for 6% of the total energy consumed (Yadav & Gahlot, 2022). Consequently, SMEs collectively have substantial

cumulative environmental impact (Alayón et al., 2022) which makes them important actors in sustainable development (Qureshi et al., 2023).

Among manufacturing industries, steel and metal manufacturing are two significant sectors from an environmental perspective as they are characterized by resource-intensive processes (Hąbek, 2025; Neri et al., 2021). Therefore, the European steel industry is aiming to reduce its emissions by 30% compared to 2018 levels (Candan & Meral, 2026) while also facing long-term climate obligations as the European Union aims to become carbon neutral by 2050 (Dragomir et al., 2022). Consequently, steel and metal manufacturing SMEs experience intense pressure and expectations in transitioning toward sustainable manufacturing (Daroń & Górska, 2019). This means that products are manufactured in a way that is not only profitable but also minimizes negative environmental impact, conserves natural resources and energy (Alayón et al., 2022) and addresses the manufacturing-related sustainability issues (Kaushal & Chakrabarti, 2023). Sustainable manufacturing considers all three dimensions of the Triple Bottom Line (TBL) as it includes economic, environmental and social perspectives when it comes to the companies' operations (Alayón et al., 2022).

Despite the growing expectation to improve environmental performance, many SMEs struggle to transform and take explicit action as they face several recurring constraints in financing, capabilities, data and measurement, and technology adaption, creating a complex landscape of barriers (Alayón et al., 2022; Neri et al., 2021). Evidence from several European SMEs show that investment hurdles, information and skill gaps, and integration challenges are reported as significant challenges in reducing negative environmental impact (Neri et al., 2021; Alayón et al., 2022). Consequently, environmental sustainability in SMEs is consistently challenged by not only external factors but also by internal limitations (Sohns et al., 2023; Yadav & Gahlot, 2022).

These challenges are intensified by specific conditions in the steel and metal manufacturing sector. Companies in this sector often rely on expensive equipment for production, operate under demanding delivery schedules, and face narrow margins, further complicating their transition towards environmental sustainability (Daroń & Górska, 2019; Goffin et al., 2024). Many of these SMEs also experience limited bargaining power due to the fact that they often operate as subcontractors of other

companies, leading to increased customer and supplier dependence and external market pressures (Neri et al., 2021; Daroń & Górska, 2019). Furthermore, environmental measurement and reporting related expectations may put additional pressure on these SMEs, while they often lack organizational capacity and effective data systems needed to meet these obligations (Hąbek, 2025; Trianni et al., 2019). Therefore, prior research shows that the environmental sustainability transition of steel and metal manufacturing SMEs operating in the EU is strongly influenced by the combined effect of environmental urgency, industrial conditions, regulatory pressures, and structural constraints (Alayón et al., 2022; Neri et al., 2021).

1.2 Problematization

SMEs have significant environmental impact, however, compared to larger companies they have received little attention in international sustainability debates (Alayón et al., 2022; Qureshi et al., 2023). A major reason for this gap is the persistent assumption of treating SMEs as smaller versions of larger companies. However, prior research highlights that SMEs should not be treated as "little big businesses" as they have different structures and characteristics compared to larger organizations (Alayón et al., 2022). SMEs often have limited financial and human resources, operate with flat organizational structures, and focus on short-term, daily operations (Alayón et al., 2022; Sohns et al., 2023). Therefore, they may face unique barriers, different from those faced by larger corporations (Alayón et al., 2022). According to Mengistu & Panizzolo (2023) and Alayón et al. (2022), these differences matter as sustainability strategies designed for larger companies cannot be scaled down without the thorough consideration of the unique conditions of SMEs.

The challenges are often exacerbated in steel and metal manufacturing SMEs operating within EU regulatory and supply chain contexts. They face increasing bureaucratic and regulatory pressures regarding their negative environmental impact, however, the fact that they face high investment costs and often lack technologies, workforce capacity, time, and access to external knowhow makes it difficult for them to transform their operations and meet these obligations (Sohns et al., 2023; Alayón et al., 2022). According to Neri et al. (2021), these SMEs often work with outdated technologies, apply weak and insufficient environmental measurement systems, and lack the capacity to hire dedicated

employees for sustainability roles. Therefore, these SMEs are in a challenging situation as they need to manage the external demands for sustainability while also focusing on their day-to-day priorities for survival (Sohns et al., 2023; Mengistu & Panizzolo, 2023).

The barriers faced by steel and metal manufacturing SMEs in their sustainable development should not be understood as isolated challenges because they often interact and reinforce each other. As an example, limited financial resources lead to constrained investments in technology, preventing SMEs from reliable environmental data collection, increasing the difficulties in identifying improvement opportunities or justifying other investments (Alayón et al., 2022; Hąbek, 2025; Neri et al., 2021). Therefore, the problem is not only the existence of several barriers, but also the way in which they jointly constrain organizational change toward reducing negative environmental impact (Alayón et al., 2022; Neri et al., 2021).

This issue is particularly important when assessed from the perspective of the Dynamic Capabilities theory. Dynamic Capabilities theory indicates that organizations that are operating in a changing environment should be able to sense emerging opportunities and threats, seize the opportunities by making timely decisions and mobilizing resources, and also transform/reconfigure its capabilities and routines over time (Teece, 2007; Bocken & Geradts, 2020). These capabilities are especially important in sustainability contexts because companies must identify environmental pressures and challenges, respond strategically, and establish technological and organizational transformations to enhance their environmental performance. (Hu et al., 2025; Zhu et al., 2025). If sustainability related barriers in steel and metal manufacturing SMEs constrain these processes, then these firms may be prevented from developing the necessary capabilities to respond effectively to challenges (Bocken & Geradts, 2020).

Although SMEs and the steel and metal manufacturing industry play an essential role in driving climate change, many of the current studies assess sustainability barriers in SMEs at a general level and they frequently lack a focused assessment of the steel and metal manufacturing industry. Furthermore, the majority of current studies are focusing on other economic sectors than the EU. Previous studies focus on the identification of barriers rather than assessing how they interact, how they reinforce each other, or how they constrain SMEs' sustainability related Dynamic Capabilities including sensing, seizing,

and transforming. Consequently, there is a need for research that assesses this specific context while providing a more integrated understanding of how barriers hinder the development of environmentally sustainable manufacturing practices in steel and metal manufacturing SMEs operating in the EU.

1.3 Purpose and Research Question

The purpose of this thesis is to explore the sector specific barriers constraining environmental sustainability in EU steel and metal manufacturing SMEs. It also examines the relationship between these barriers and sustainability related Dynamic Capabilities. More specifically, this thesis aims beyond identifying the internal and external barriers by also analyzing how these barriers affect the studied companies' ability to sense environmental sustainability related challenges and opportunities, seize appropriate strategic and operational responses, and transform their existing practices and organizational routines (Teece, 2007; Bocken & Geradts, 2020). This study addresses the gaps found in current literature and aims to contribute to a more theoretically informed and deeper understanding of the reasons behind the difficulties faced by the studied SMEs in reaching their environmental sustainability targets. Therefore, the research question of this study is:

How do barriers constrain the environmental sustainability of EU steel and metal manufacturing SMEs in relation to sensing, seizing, and transforming capabilities?

1.4 Delimitations

This thesis study is empirically delimited to Hungarian steel and metal manufacturing SMEs. The institutional, regulatory, and supply chain levels of this study are European Union level, while the empirical data is from Hungarian SMEs. However, the observed companies are embedded in international, EU-wide supply chains as they are sourcing materials from outside of Hungary and they are supplying products to other EU countries. The assessed companies also focus on EU sustainability regulations, making the discussed barriers relevant to the European Union context. This study does not aim for the generalization of the findings in large populations, it rather seeks to provide analytical insight on the barriers faced by the studied steel and metal manufacturing SMEs.

In many cases, sustainability is explained through the Triple Bottom Line (TBL) concept, which includes economic, environmental and social dimensions (Alayón et al., 2022). However, this thesis is delimited to study only the environmental dimension. Social and economic dimensions are still treated as relevant, but narrowing down the scope is necessary to effectively examine the barriers to environmental impact reduction, while ensuring that the main goal and focus of the study is not overshadowed by less central dimensions. The thesis is further delimited to examining the barriers to environmental sustainability meaning that our study does not aim to identify and analyze the potential enablers of sustainable development within the studied SMEs.

2. Frame of Reference

The frame of reference section introduces the literature selection process. It further elaborates the concepts and theories introduced in the previous section. The topics that are covered involve sustainable manufacturing, sustainability reporting, circular economy, a taxonomy of barriers hindering change and sustainable development, and the dynamic capabilities theory for sustainability as an exploratory lens. Lastly, the section also discusses the gaps in current literature.

2.1 Method of Constructing the Frame of Reference

Finding the appropriate literature for the frame of reference section involved several steps. Our aim was to select, assess, and synthesize the most relevant studies regarding the barriers to environmental sustainability in manufacturing SMEs, particularly, steel and metal manufacturing SMEs as well as the theoretical perspective of our thesis. Therefore, a structured approach was applied for the literature selection process to increase transparency.

After deciding on the topic, aim, and research question of our thesis, we conducted a comprehensive search using Scopus, one of the biggest scholarly databases for peer-reviewed literature (Gil et al., 2024; Sauer & Seuring, 2023), using keywords that we identified based on the characteristics of the studies that we needed. We identified 4 groups of keywords. The first group focused on the barriers and challenges, the second group focused on sustainability and sustainable development, the third group focused on manufacturing, and the fourth group focused on SMEs. In the designed search query, the "AND" and "OR" Boolean operators (Kraus et al., 2022) were also applied to capture the intersection of these different keyword groups (Sauer & Seuring, 2023). The last important element of our search query was using truncation at the end of several keywords. Using these truncations allowed us to capture all sets of spelling variations, consequently not losing potentially relevant literature (Kraus et al., 2022). Overall, this keyword search resulted in 654 entries in our preliminary sample.

Table 1 Search Query Used in Scopus Keyword Search

Search query used in Scopus
(TITLE-ABS-KEY (barrier*) OR TITLE-ABS-KEY (challenge*) AND TITLE-ABS-KEY (sustainab*) OR TITLE-ABS-KEY (sustainable development) AND TITLE-ABS-KEY (manufactur*) AND TITLE-ABS-KEY (Small AND medium enterprise*) OR TITLE-ABS-KEY (small AND medium-sized enterprise*) OR TITLE-ABS-KEY (SME) OR TITLE-ABS-KEY (SMEs))

The next step of our literature selection process was to refine the first sample of 654 papers by applying several inclusion categories available within the Scopus database. We used filters in three main areas: Subject Area, Document Type, and Language. We restricted the results to "Business, Management and Accounting", "Economics, Econometrics and Finance", Environmental Sciences", and "Social Sciences" to ensure disciplinary relevance (Kraus et al., 2022). Our review excludes editorials and conference papers as they are often not checked for quality or they are less mature (Kraus et al., 2022; Sauer & Seuring, 2023). Therefore, we applied a filter to only include journal articles, the most "valuable sources" from research perspectives increasing transparency (Kraus et al., 2020). However, this criterion can also be a limitation as it excludes some potentially valuable documents, but we were able to still collect an adequate amount of reliable journal articles for our review (Sauer & Seuring, 2023; Gil et al., 2024). Lastly, we applied the third filter to limit the search results to articles written in English as articles written in another language are not available to a worldwide research community (Gil et al., 2024). Applying these filters led to the exclusion of 389 records, leaving 265 articles for further title, keyword, and abstract screening.

The sample of 265 articles was exported to an Excel table where we conducted a structured screening process reading through the titles, abstracts, and author-defined keywords as suggested by Hiebl (2021). This phase resulted in the further exclusion of 178 articles because they did not focus on barriers to environmental sustainability, emission reduction, or did not deal with the manufacturing sector.

The remaining 87 articles were assessed for final eligibility through reading the full texts. A further 62 articles were removed in this phase as they failed to meet the specific industrial or geographic scope of this paper. Specifically, the excluded articles did not focus on SMEs in European Union countries or with the steel manufacturing sector. Therefore, the manual screening process including abstract, keyword, title, and full-text analyses led to the inclusion of 25 full-text articles selected from the preliminary database.

As Hiebl (2021) highlights, keyword searches can occasionally miss relevant and valuable articles. Therefore, we supplemented our Scopus database search by conducting a manual search of additional literature (Gil et al., 2024) in Primo Library, the online library database of Jönköping University. We also used citation tracking in the documents identified in the Scopus database search. These additional steps led to the inclusion of 13 articles bringing the final review sample to 38 journal articles.



Figure 1 PRISMA Flowchart of the Literature Selection Process

To maintain high scientific standards and assure reliability we conducted a quality assurance process. We selected articles that were published in journals that received rankings in the Australian Business Deans Council (ABDC) and Scopus journal ranking engines. We selected the final sample of 38 articles following this quality criterion as it ensures that the review section builds on peer reviewed and relevant empirical evidence regarding the barriers to the sustainable development of European SMEs and steel manufacturing SMEs.

While some guidelines in prior studies recommend larger samples for a literature review, a lower number of literature can be considered appropriate in highly specialized or emergent topics (Hiebl, 2021). Therefore, as our research is focusing specifically on SMEs operating in the EU steel and metal manufacturing sector, the number of articles assessed in the review section is considered as justified for this specialized topic.

2.2 Environmental sustainability in manufacturing SMEs

As this thesis is informed by literature on the environmental sustainability of steel and metal manufacturing SMEs and European manufacturing SMEs, this thesis uses the European Commission's SME definition: ‘enterprises with less than 250 employees and an annual turnover less than €50 million (or a balance sheet lower than €43 million)’ (European Commission, n.d.; Alayón et al., 2022).

On the other hand, environmental sustainability is understood in this thesis as the ability of a manufacturing firm to reduce its negative impact on the environment, while also being able to produce and create value. This includes lowering emissions, reducing waste and the amount of material they use, improving efficiency and using more responsible processes throughout the production (Kaushal & Chakrabarti, 2023; Trianni et al., 2019). Therefore, environmental sustainability is not treated as an abstract goal, but as a combination of actions that affects a manufacturer’s environmental footprint.

To take the ‘first step’ towards being sustainable, moving from traditional manufacturing practices to a new approach that integrates environmental (and social) perspectives is presented as important (Mengistu & Panizzolo, 2023). To connect environmental sustainability and manufacturing, some of the analyzed literature uses the ‘sustainable manufacturing’ framework; finding a different production approach, which reduces

environmental harm, cuts down energy use and resource consumption, and keeps people safe, including workers, customers, and surrounding communities (Alayón et al., 2022; Kaushal & Chakrabarti, 2023). To achieve such development and reduce negative environmental impact, it is important that the manufacturing systems are designed in a restorative and regenerative way from the start (Howard et al., 2022). Therefore, sustainable manufacturing is closely connected to the Circular Economy concept, which is a restorative and regenerative business model design, focusing on minimizing the amount of resources needed for production through processes like recycling, remanufacturing, and repairing, thus, minimizing waste, resource consumption, and reducing negative environmental impact (Howard et al., 2022). Combining these models with technological advancements that drive the companies' transition toward more sustainable production, support their efforts to reduce emissions and limit their overall environmental impact (Olipp et al., 2025; Abdulnour et al., 2022).

2.3 Sustainable manufacturing and environmental transition

In the steel and metal manufacturing industry, sustainable manufacturing is especially difficult because of the structural characteristics of the sector. These industries are highly resource- and energy-intensive and depend on continuous production, which makes reducing environmental impacts particularly complex and difficult in this sector (Hąbek, 2025; Neri et al., 2021). At the same time, regulatory expectations and EU targets are increasing the pressure on the companies in this sector (Candan & Meral, 2026; Dragomir et al., 2022). This creates a clear tension: the urgency is growing, but the operational reality still restricts how fast the SMEs can respond (Daroń & Górska, 2019).

A key characteristic of this sector is that it is highly energy intensive, and reliant on expensive, special machines. Installing these machines and systems require substantial upfront investments, and once they are operating, replacing or upgrading them is challenging. This makes it difficult for firms to switch to cleaner technologies. In addition, many SMEs are still working with old, outdated machines, but their financial situation and the risk of failure itself do not encourage trying new solutions or technologies. (Daroń & Górska, 2019; Pech & Vrchota, 2020)

Another aspect that defines this sector is how much SMEs rely on subcontracting. Many of them work as suppliers to significantly larger companies, meaning their bargaining

power is limited and they depend heavily on external partners (Daroń & Górska, 2019; Neri et al., 2021). They have to follow strict rules on for example cost and quality, reducing the space for them to invest in their own sustainability strategies. As a result, due to client expectations, they tend to focus on efficiency and cost reduction instead of long-term sustainability innovations (Daroń & Górska, 2019; Pech & Vrchota, 2020).

Furthermore, SMEs in this sector are not only smaller than larger firms, but they are also structurally different and that makes sustainability transitions much harder (Alayón et al., 2022). They usually operate with limited financial and human resources, lack in-house sustainability expertise, and often lack dedicated systems for environmental monitoring and reporting (Alayón et al., 2022; Neri et al., 2021). Unlike large corporations that can invest in advanced technologies, SMEs are often focused on surviving day-to-day operations, thus, sustainability is not a top priority (Sohns et al., 2023; Mengistu & Panizzolo, 2023). This supports the argument that strategies designed for such large firms cannot be directly applied to SMEs (Alayón et al., 2022).

In addition, SMEs in this sector are under increasing pressure in relation to reporting. They are expected to track and report data, but many simply do not have tools to do so (Hąbek, 2025; Trianni et al., 2019). With weak data systems, limited digitalization, and fragmented processes, it becomes difficult to understand their own environmental impact (Hansen et al., 2025; Müller et al., 2024). This not only slows down their sustainability transition but also reinforces the barriers already constraining their transition.

Closely connected to this reporting pressure is ‘twin transition’. Companies face expectations to become both digital and sustainable at the same time (Ingaldi & Ulewicz, 2025; Muruganantham et al., 2025). Tools like IoT and Big Data are increasingly important as they are essential for tracking emissions, managing waste, and improving energy efficiency (Abdel-Aty et al., 2024; Ingaldi & Ulewicz, 2025). However, SMEs may struggle because their actions are limited by their digital maturity, lack of financial resources, infrastructure, and skilled workforce needed to adopt these smart manufacturing technologies (Hansen et al., 2025; Ingaldi & Ulewicz, 2025). For example, many are still close to ‘level 0’, meaning they have little to no understanding of how digital tools could support their operations (Hansen et al., 2025). Outdated software and incompatible systems remain in use, making tracking energy or material usage hard

(Hansen et al., 2025; Müller et al., 2024). In addition, managers tend to be hesitant to invest in new digital solutions due to high costs and concerns about cybersecurity risks (Ingaldi & Ulewicz, 2025; Müller et al., 2024). Overall, this urgency for sustainability and the difficulties in transitioning cause a complex mix of barriers that may affect SMEs' environmental sustainability transition.

Taken together, environmental sustainability is constrained by a double pressure in steel and metal manufacturing SMEs. Firstly, companies face emerging external expectations from customers, supply chain partners, and regulatory bodies (Hąbek, 2025). Secondly, their internal capacities to address these expectations are limited due to resource constraints, outdated technologies, weak data systems, and operational priorities focusing on short-term goals (Alayón et al., 2022; Neri et al., 2021; Hansen et al., 2025; Müller et al., 2024). This tension is central for this thesis because it indicates that the sustainability barriers faced by these SMEs are interconnected, forming a system of barriers.

2.4 Taxonomy of barriers hindering change

This complex mix of barriers is presented here (Table 2) as a taxonomy of barriers that SMEs face in their sustainable transition. The revised literature categorizes barriers into internal and external dimensions, emphasizing the importance of distinguishing between their origins and the degree of control SMEs have over them. Internal barriers refer to factors that originate within the organization and directly influence the adoption of sustainable manufacturing practices, which SMEs may actively manage and control. On the contrary, external barriers occur outside the organization and affect the adoption of sustainability practices through influencing factors that SMEs have no direct control over. (Alayón et al., 2022)

Table 2 Internal and External Barriers to Sustainable Development in SMEs

Barrier Type	Category	Specific Challenges	Key References
Internal	Financial & Resource Constraints	High initial investment, limited financial capital, limited resources, uncertain ROI, hidden regulatory costs	Abdel-Aty et al., 2024; Abdulnour et al., 2022; Alayón et al., 2022; Goel et al., 2022; Hąbek, 2025; Muruganantham et al., 2025; Bhavani et al., 2023; Neri et al., 2021; Sabale et al., 2025
Internal	Knowledge, Skills & Expertise	Shortage of skilled employees, lack of sustainability know-how, lack of digital skills for Industry 4.0, difficulty implementing LCA, circular design, and double materiality	Abdel-Aty et al., 2024; Hąbek, 2025; Alayón et al., 2022; Bagagiolo et al., 2023; Daroń & Górska, 2019; Ingaldi & Ulewicz, 2025; Muruganantham et al., 2025; Müller et al., 2024; Kaushal & Chakrabarti, 2023; Sabale et al., 2025
Internal	Organizational Culture & Management Commitment	Resistance to change, weak strategic direction, insufficient top management commitment, focus on day-to-day operations	Daroń & Górska, 2019; Ingaldi & Ulewicz, 2025; Muruganantham et al., 2025; Alayón et al., 2022; Hansen et al., 2025; Neri et al., 2021; Sabale et al., 2025; Sohns et al., 2023
Internal	Data & Technological Infrastructure	Aging systems, isolated automation, lack of real-time monitoring, insufficient data infrastructure for sustainability	Hansen et al., 2025; Alayón et al., 2022; Hąbek, 2025; Ingaldi & Ulewicz, 2025; Kaushal & Chakrabarti, 2023; Olipp et al., 2025; Goffin et al., 2024; Müller et al., 2024
External	Regulatory Complexity & Bureaucracy	Complex/changing regulations, heavy paperwork, unclear policies, inconsistent standards, lack of funding	Alayón et al., 2022; Hąbek, 2025; Muruganantham et al., 2025; Neri et al., 2021; Olipp et al., 2025; Dragomir et al., 2022; Ingaldi & Ulewicz, 2025; Isensee et al., 2023; Müller et al., 2024; Sabale et al., 2025
External	Market Pressure & Weak Customer Demand	Price-sensitive markets, hesitant consumers, uncertain/noisy demand for sustainable products	Neri et al., 2021; Alayón et al., 2022; Bhavani et al., 2023; Qureshi et al., 2023; Muruganantham et al., 2025; Bagagiolo et al., 2023; Zhu et al., 2025
External	Supply Chain & Ecosystem Limitations	Weak collaboration, lack of circular economy integration, reliance on larger partners, limited bargaining power	Bhavani et al., 2023; Hąbek, 2025; Hansen et al., 2025; Howard et al., 2022; Sohns et al., 2023
External	Insufficient External Infrastructure	Lack of recycling/collection hubs, shared data platforms, EV charging stations, logistical constraints	Müller et al., 2024; Sabale et al., 2025; Howard et al., 2022

However, the differentiation between the internal and external origins of the barriers should not imply that they constrain the SMEs' sustainable development independently. It rather indicates an analytical distinction that helps understand and explain the interaction between these barriers. As assessed in section 2.3, they are closely connected as internal barriers can reduce the SMEs' ability to respond to external events and expectations while external barriers can further intensify the effects of internal barriers (Alayón et al, 2022; Neri et al., 2021). For example, market pressure for sustainable development can increase the demand for consistently measured and reported environmental data, however, SMEs' weak internal data systems and limited expertise makes it challenging to respond effectively (Hąbek, 2025; Hansen et al., 2025; Müller et al., 2024). Consequently, environmental sustainability-related barriers are not isolated obstacles, and it is important to understand how they combine and reinforce each other. Sustainability reporting has a major role in establishing this connection and reinforcement between the internal and external barriers, while the Dynamic Capabilities theory helps examine how these barriers constrain the companies' ability to respond to the challenges.

2.5 The Interface Between Internal and External Barriers: Sustainability Reporting

Sustainability reporting has become an important area in which the broader sustainability transition is placing pressure on steel and metal manufacturing SMEs (Kozłowska &

Ustinovičius, 2025; Hąbek, 2025). Requirements for CSRD-related statements in combination with increasing value-chain expectations create a rapidly growing need for firms to provide detailed, verifiable, and comparable data regarding their sustainability activities, (particularly in relation to Scope 3 emissions) (Hąbek, 2025; Martinčević et al., 2025). For many manufacturing firms, this means a major shift, as reporting requires auditable data that can be communicated to customers, regulators, and other stakeholders (Abdel-Aty et al., 2024; Hąbek, 2025). In this sense, reporting is becoming not only an administrative task, but also increasingly important for remaining included in larger supply chains and markets (Abdel-Aty et al., 2024; Martinčević et al., 2025).

For steel and metal manufacturing SMEs, this challenge is difficult to overcome since sustainability reporting sits where internal and external barriers meet (Alayón et al., 2022; Hąbek, 2025). On one hand, firms face internal difficulties such as weak digital systems, poor quality of data, fragmented processes, and limited organizational capacity to gather and manage crucial information (Hansen et al., 2025; Hąbek, 2025; Müller et al., 2024). On the other hand, reporting requirements are increasingly shaped by external factors such as regulators, customers, and supply-chain partners, who expect more transparency and more standardized environmental data (Martinčević et al., 2025; Rantala et al., 2025). This tension is notably visible regarding Scope 3 emissions. For many manufacturers Scope 3 emissions often represent the largest portion of the total environmental footprint, yet they are difficult to measure as they depend heavily on external data (Hąbek, 2025; Rantala et al., 2025). Thus, reporting can become one of the clearest points at which internal weaknesses and external pressures directly collide (Hansen et al., 2025; Martinčević et al., 2025).

For credible reporting, firms should not only identify relevant environmental information, but also embed more structured monitoring into daily operations, improve system integration, and coordinate more effectively with supply-chain partners (De Almeida Barbosa Franco et al., 2024; Hąbek, 2025; Martinčević et al., 2025). Where internal processes remain fragmented and external data flows remain weak, reporting is likely to be unreliable or incomplete and inconsistent over time (Hansen et al., 2025; Müller et al., 2024; Rantala et al., 2025). In this sense, reporting shows how sustainability transition is shaped not just by barriers, but by the collision between external pressure and limited internal capabilities available to respond to them (Alayón et al., 2022; Zhu et al., 2025).

Sustainability reporting can therefore be understood as a key interface between the internal and external barriers (Alayón et al., 2022; Hąbek, 2025).

2.6 Theory: Dynamic Capabilities as explanatory lens

As presented above, separating barriers into internal and external categories helps to organize the challenges faced by SMEs in the steel and metal manufacturing industry. It is also found that sustainability reporting can act as the interface that connects external and internal barriers. (Alayón et al., 2022; Hąbek, 2025)

However, they do not fully explain how these barriers affect SMEs' ability to respond to sustainability-related pressures. They show what the barriers are and how they connect but do not fully explain how they limit organizational actions. Dynamic Capabilities theory acts as a tool for interpreting how barriers may affect the responses of companies. Instead of treating barriers as obstacles on their own, the Dynamic Capabilities perspective helps to explore and understand how they hinder the SMEs' sustainability transition. More specifically, how the barriers constrain sensing, seizing, and transforming in the companies. (Teece, 2007; Bocken & Geradts, 2020; Zhu et al., 2025)

Sensing capabilities refer to a firm's ability to 'read the environment', including the identification of risks and opportunities for example understanding regulations or following market trends (Bocken & Geradts, 2020; Teece, 2007). For the SMEs in question, this capability is often weakened by internal barriers such as limited data systems, lack of expertise, and limited human resources (Hąbek, 2025; Hansen et al., 2025; Müller et al., 2024; Alayón et al., 2022). On the other hand, outdated and fragmented systems and limited knowledge make it harder for SMEs to implement tools like CSRD or apply tools like Life Cycle Assessment (Goffin et al., 2024; Kaushal & Chakrabarti, 2023). External factors, including complex and unclear regulations also increase uncertainty and confusion (Alayón et al., 2022; Neri et al., 2021; Isensee et al., 2023). This suggests that limited sensing capabilities do not necessarily mean that SMEs are ignoring sustainability, but may instead indicate limited access to reliable information, weak data systems, and uncertainty in sustainability interpretation.

Seizing capabilities are about turning awareness into action, making decisions and actually investing in solutions once opportunities have been identified (Teece, 2007;

Bocken & Geradts, 2020). However, for SMEs, this is where implementation is frequently constrained. A central reason identified in the literature is financial pressure. Sustainability investments require high upfront investment, and their short-term financial returns make it difficult to justify them (Alayón et al., 2022; Neri et al., 2021; Bhavani et al., 2023; Goel et al., 2022). The hesitation is reinforced by unclear market demand, as customers are not always willing to pay more for sustainable products (Muruganantham et al., 2025; Qureshi et al., 2023). At the same time, these SMEs operate in highly competitive markets, where short-term survival is often the top priority (Sohns et al., 2023; Mengistu & Panizzolo, 2023). Managers may also avoid risky decisions due to time pressure and uncertainty (Alayón et al., 2022; Sabale et al., 2025). As a result, SMEs may recognize sustainability opportunities but still decide not to act on them.

Transforming capabilities refer to the reconfiguration of how a company operates by updating technologies, processes, and organizational routines (Teece, 2007; Bocken & Geradts, 2020). In the context of this thesis, transforming involves moving beyond small sustainable manufacturing processes and actually redesigning how companies operate. This step is often presented as the most challenging. Many SMEs rely on expensive, special machines, which makes switching to new technologies challenging (Daroń & Górska, 2019; Pech & Vrchota, 2020). Internal resistance to change and limited management commitment can further set back this process (Ingaldi & Ulewicz, 2025; Hansen et al., 2025; Alayón et al., 2022). Limited supply chain integration and dependence on mostly larger partners further restrict systemic changes (Bhavani et al., 2023; Howard et al., 2022; Sohns et al., 2023), and weak external infrastructure adds to these constraints (Müller et al., 2024; Sabale et al., 2025). Consequently, many SMEs struggle to move beyond their existing systems, even when they are aware of the need for change.

Overall, the Dynamic Capabilities framework shows that the mentioned barriers are interconnected and affect many actions and that sustainability challenges are not just about limited resources, they are about limited capabilities that prevent SMEs from moving forward (Teece, 2007; Bocken & Geradts, 2020; Zhu et al., 2025).

2.7 Gaps in the Current Literature

As assessed through the frame of reference section, prior research has generated important insights and understanding on sustainability barriers faced by SMEs in the manufacturing industry within and also outside of the EU. There are existing studies that focus specifically on the EU steel and metal manufacturing sector when examining these aspects, however, the majority of the research tend to assess a broader scope, mostly examining SMEs across different sectors, geographical locations, or thematic lenses including sustainable manufacturing (Alayón et al., 2022), circular economy (Howard et al., 2022; Olipp et al., 2025), digitalization and industry 4.0 (Abdel-Aty et al., 2024; Ingaldi & Ulewicz, 2025; Müller et al., 2024) and sustainability reporting (Hąbek, 2025; Trianni et al., 2019). Overall, the literature presents limited analysis of the barriers to environmentally sustainable development in the specific sector of steel and metal manufacturing SMEs in countries within the European Union.

Furthermore, there is also an existing theoretical gap in the current assessment of the environmental sustainability related barriers, concerning the application of the Dynamic Capabilities theory which has become an important framework in understanding companies' ability to adapt in changing environments (Bocken & Geradts, 2020; Teece, 2007). As indicated in the previous section, assessing the sustainability barriers through the lens of the Dynamic Capabilities theory offers a deeper understanding on the internal and external barriers faced by SMEs in their sustainable development. There are studies that assess the role of Dynamic Capabilities in sustainability-related contexts, emphasizing their supporting role in responding to environmental challenges and developing towards environmentally sustainable business operations. However, the current body of literature focuses mostly on either the identification of barriers or the assessment of the companies' Dynamic Capabilities rather than the relationship between these aspects. In other words, there is only a limited amount of existing research focusing on how the barriers relate to the Dynamic Capabilities needed for SMEs to be able to respond to environmental sustainability related opportunities and threats, and thus, establish environmentally sustainable manufacturing operations. This theoretical gap is particularly relevant in the context of this thesis as based on several studies, the sustainability transition of steel and metal manufacturing SMEs depend not only on external obligations and pressures but also on internal capabilities and their ability to

change their organizational processes, aligning with environmental sustainability perspectives. (Daroń & Górska, 2019; Goffin et al., 2024; Bocken & Geradts, 2020)

Taken together, based on the gaps identified above, the currently available literature remains fragmented for understanding the complexity of the barriers constraining environmental sustainability in relation to Dynamic Capabilities within European regulatory and supply chain contexts. Therefore, there is a need for further studies that conduct a context-specific and more integrative assessment of the barriers and their interconnectedness while applying the Dynamic Capabilities theory as an exploratory lens.

3 Methodology

This section first introduces the research paradigm and approach of the thesis. Following, it further discusses the research design, introducing how data were collected for the study. It also elaborates on how the collected data were analyzed. Lastly, this section also discloses the ethical considerations and trustworthiness of the study as well as the limitations of the method used.

3.1 Research paradigm

Academic literature distinguishes between two common research paradigms, interpretivism and positivism. According to Lee (1991), these two paradigms are differentiated based on whether the study is objective or subjective and qualitative or quantitative. When it comes to positivist studies, this term refers to the procedures of inferential statistics, hypothesis testing, or mathematical analyses (Lee, 1991). On the other hand, interpretivist approach is applied to case studies or phenomenology (Lee 1991). In other words, interpretive research focuses on meanings, contexts, and understanding while positivist research focuses on explanation, measurement, and generalization (Bonache & Festing, 2020). This study aims to reveal how company representatives identify and understand the context-sensitive sustainability barriers of their companies. Therefore, this research must consider that every company is different, making interpretivist research the suitable and more appropriate paradigm, given the fact that this study aims for understanding the perception of the people in the given "reality" rather than generalizing and applying a statistical testing of previously defined relationships (Bonache & Festing, 2020).

3.2 Research approach

The two most common research approaches are the inductive and deductive approaches. According to Woiceshyn & Daellenbach (2018), the two approaches follow different logics as deduction involves that the study starts from a theory which is used to form hypotheses, then those hypotheses are tested, and the theory is revised. On the other hand, when it comes to inductive approach, the study starts with empirical observations and

works towards identifying patterns, themes, and formulating a broader conceptual understanding using the gathered data (Woiceshyn & Daellenbach 2018).

In this study the literature review introduces and assesses the academic and theoretical background of the research topic. Then the study builds on the Dynamic Capabilities theory and the findings of previous literature regarding the barriers to environmental sustainability in manufacturing SMEs. However, as indicated in the previous section, there is limited framework focusing on the relationships between environmental sustainability barriers and the Dynamic Capabilities theory, especially in the specific industry of steel and metal manufacturing SMEs. Therefore, a deductive approach with hypothesis testing would be premature. Instead, this study uses the Dynamic Capabilities theory and previously identified barriers as exploratory lenses to allow new empirical evidence and themes to emerge. Therefore, this study applies an abductive approach which merges the deductive and inductive approaches (Hurley et al., 2021). This abductive logic was adopted as it enables using previous findings and theory while also remaining open to new insights and themes derived from the empirical study that may not align with the existing barrier frameworks (Hurley et al, 2021). Overall, this approach allows the study to combine theory-driven expectations with new empirical observations to provide the most plausible explanations (Hurley et al., 2021).

3.3 Method of Data Collection

Given its purpose, this study is qualitative in nature and for data collection it uses semi-structured interviews with company representatives. According to Lim (2025), using interviews for qualitative research purposes is particularly appropriate when the aim of the study is to assess a specific context or to explore the experiences or perspectives of the participants.

The nature and the purpose of this study also motivated the decision of using semi-structured interviews instead of structured or unstructured interviews. According to DiCicco-Bloom & Crabtree (2006), structured interviews tend to generate quantitative data, therefore, this data collection method was ruled out from this study to eliminate the risk of restricting the participants in explaining the context specific barriers faced by their companies. Unstructured interviews offer a deep understanding of themes without the use of an interview guide, however, they tend to widen and expand the initial scope of the

study (Lim, 2025). Semi-structured interviews are somewhere between the structured and unstructured interviews. They use an interview guide which guides the interviewer during the interview allowing to explore suggested aspects, issues, and questions while also offering a level of flexibility to follow emerging themes and form a deeper understanding of the topics (Lim, 2025). The researchers wanted to keep the initial scope of the study, focusing purely on the environmental sustainability-related barriers faced by the assessed SMEs while considering the Dynamic Capabilities theory. On the other hand, the aim of the study is to form a deep understanding of these aspects based on the experiences and insights of the participants. For this, allowing themes and aspects to unfold during the interviews is essential. Therefore, semi-structured interviews were considered as the most appropriate method for data collection, providing a mix of structure and flexibility. To collect data, four semi-structured interviews were conducted with the representatives of four different Hungarian steel and metal manufacturing SMEs that are embedded in EU-wide supply chains and are exposed to EU rules and regulations. These interviews yielded 288 minutes of data.

3.3.1 Sampling Process

Purposeful sampling was adopted through the research, to identify the most appropriate companies and interviewees to participate in this study as it allowed for the inclusion of cases that relate to the research question (Stratton, 2024). The SMEs were selected depending on their relevancy to the research subject in terms of sustainability in their operations, geographical location, and their capacity to practice sustainability in the industry of steel and metal manufacturing. It was an essential aspect that the SMEs must be headquartered in the European Union. The researchers also considered several aspects when selecting companies based on the participants from the assessed SMEs. It was an essential inclusion criterion that the interview participants from these firms held positions that would provide them with information and experience regarding sustainability. Overall, these actions guided the researchers in identifying relevant, potential participants. (Malterud et al., 2015; Vasileiou et al., 2018)

After extensive research, 21 EU steel and metal manufacturing SMEs were identified as appropriate candidates for this thesis study. These companies offered a very strong initial sample of several EU countries including Sweden, Italy, Poland, Hungary, and Slovakia. Companies were initially contacted through email in which the researchers kindly asked

for their participation. For the first emails, only 2 companies replied ensuring participation while 11 other companies declined the request. Another round of emails was sent to the remaining companies which yielded one more participant. The researchers also received 5 rejections. In the third and last round of emails 1 more company agreed to participate in this research study.

3.3.2 Sample

The sample of this study consists of four Hungarian steel and metal manufacturing SMEs that are operating internationally. Although the aim of the sampling process was to include companies that are based in different EU countries, the empirical sample became concentrated geographically to Hungarian companies due to access limitations and the SMEs' willingness to participate. Despite the fact that every SME in this study is based in Hungary, each of them is operating internationally, supplying to other EU countries and sourcing their raw materials internationally. They are exposed to EU standards and international customer requirements, consequently, the sample provides valuable insights on how steel and metal manufacturing SMEs operating in the European Union experience barriers in their sustainable development. Table 3 summarizes the final sample.

Table 3 Final Interview Sample

Company	Number of Employees	Country	Position of Participant	Interview Length
Company 1	85	Hungary	CEO	92 minutes
Company 2	131	Hungary	CEO / Member of the Executive Board	82 minutes
Company 3	107	Hungary	CEO / Owner	49 minutes
Company 4	56	Hungary	CEO / Owner	65 minutes

3.3.3 Interview Guide

Before selecting and contacting the potential participants and conducting the interviews, an interview guide was developed to support the semi-structured interviews. According to Kallio et al. (2016), semi-structured interviews are useful when the aim is to explore the participants' perceptions, experiences, and opinions, while still maintaining a clear connection to the core purpose of the research. The interview guide was a helpful tool to ensure this connection is maintained.

In line with Kallio et al. (2016), this interview guide was built around the main themes that are relevant for this study. Each theme contained 3 or 4 predetermined questions allowing the researchers to elaborate on the most important aspects regarding the purpose of the study. The first theme of the interview guide plays an essential role in the interviews. According to DiCicco-Bloom & Crabtree (2006), developing a so-called "rapport" (trust, respect) is especially important in semi-structured interviews to be successful. Therefore, the first theme of the interview guide is important as it acts as a conversation starter, helping the researchers in establishing an environment in which the participants feel respected, safe, and comfortable. The other themes of the guide explore the most important aspects of the topic based on the previously reviewed literature in Section 2. This follows Kallio et al. (2016), who argue that previous knowledge should be used as a basis for developing the interview guide, as it helps create a conceptual foundation for the interview and ensures that the questions are connected to the study purpose. The themes of the interview guide in order are:

1. Introduction and background
2. Current sustainability practices
3. Barriers and challenges
4. Internal capabilities and infrastructure
5. Sensing
6. Seizing
7. Transforming
8. Closing

The first half of the themes focus on understanding the companies' current approach to sustainability while exploring the main challenges they face regarding sustainability. This

section also assesses the current internal capabilities of the companies involving their knowhow, infrastructure, data management, and technological support. Therefore, the first half of the themes build on the findings of prior literature assessed in the literature review section. The second half of the themes focus on the dimensions of the Dynamic Capabilities theory by exploring the companies' ability to identify sustainability-related opportunities or threats, their ability to seize these opportunities or eliminate threats, and their ability to transform and adapt their processes and operations accordingly. The structure of the guide also reflects methodological recommendations indicating that interview guides should be organized thematically and should move from more general and less sensitive topics toward more specific and reflective questions (Abraham & Padmakumari, 2026). The interview guide ends with a closing theme in which a closing question allows participants to freely reflect on what would help their company in its sustainability transition. (Appendix 1)

3.3.4 Recording and Transcription

Recording an interview is an important part of a qualitative study which uses semi-structured interviews as its data collection method. It allows the researchers to avoid having to divide their focus between the interview and taking notes by hand, therefore, engaging fully with both the participant and the conversation (DiCicco-Bloom & Crabtree, 2006; Lim, 2025).

The interviews were conducted online, in English, using Microsoft Teams. This online method was the most appropriate format for this thesis because it allowed better coordination and enabled efficient access to the company representatives despite the limited resource and time constraints of this study. It also allowed the interviews to be recorded and transcribed using the real-time transcription engine of the Microsoft Teams software. The written transcriptions were then reviewed manually to take out every personal and identifying detail to keep anonymity, then scanned with limited AI assistance to find spelling and grammar errors. This approach allowed the researchers to transform the information and data collected during the interviews into an analyzable form.

3.4 Data Analysis

The representatives of four Hungarian steel and metal manufacturing SMEs were interviewed through detailed conversations. Semi-structured interviews conducted online served as the empirical material for the analysis. The data were examined using a thematic coding approach grounded in a hybrid deductive–inductive logic meaning that the codes were partly informed by existing theoretical frameworks, while also allowing new themes to emerge directly from the data (Fereday & Muir-Cochrane, 2006). Thematic analysis was selected because it allows the analysis of complex material without imposing an entirely predetermined shape (allows the research to honor both established frameworks and unexpected findings that only emerge in an empirical setting) (Braun & Clarke, 2006). Based on the abductive research approach of this study, the hybrid deductive-inductive thematic coding is appropriate as it includes existing theory while remaining open for new and emergent themes and observations.

3.4.1 Stepwise analytic procedure

Every conversation was captured in full, audio recorded with consent, and transcribed verbatim. The interviews were initially transcribed with Microsoft Teams, then the transcripts were anonymized by removing any identifier details. The anonymized transcripts were then revised both manually and with limited AI assistance to find spelling and grammar errors. The AI tool was not used to generate interpretations, codes, themes, or findings, and every suggested correction was checked manually by the authors to ensure that the meanings remained the same. Then the detected errors were corrected manually. Interviews were transcribed verbatim to preserve meaning and support rigorous coding. The transcripts were read repeatedly to deepen the familiarity with the empirical material. The initial impressions and emerging patterns were noted and later developed as codes (Nowell et al., 2017).

Before starting to code any of the transcripts, the literature provided a clear analytical direction. The aggregates of the preliminary codebook were established based on existing research and the Dynamic Capabilities theory. They were organized around categories that already had strong theoretical support behind them, such as internal barriers, external barriers, and the three dimensions of Dynamic Capabilities which are sensing, seizing, and transforming (Alayón et al., 2022; Teece, 2007). These categories provided the initial

coding structure, ensuring consistency and comparability across all four interviews from the beginning (Fereday & Muir-Cochrane, 2006). Based on the reviewed literature and the Dynamic Capabilities theory, six aggregates were defined: internal barriers, external barriers, interface of internal and external barriers, sensing, seizing, and transforming. During the coding process, these six aggregates were refined into 13 themes and 30 concepts, based on empirical patterns in the interviews. The codebook was color coded by each aggregate and every concept was labeled with a number. (The initial codebook is found in Appendix 3).

Each transcript was then color coded manually. Each coded line was organized in an Excel table and labelled with the appropriate number for further clarity. In instances when a respondent described a barrier or relationship that did not fit within any pre-established category but clearly defined relevant barriers, a new code was generated. This approach allowed the coding framework to remain responsive rather than imposing a fixed, predetermined structure onto the data, thereby preserving the analytical integrity of participants lived experiences and conceptual distinctions. These emerging codes were documented carefully and added into the codebook as emergent codes, making sure that relevant or unexpected patterns coming directly from the data were assigned to an appropriate analytical position in the codebook and analysis (Fereday & Muir-Cochrane, 2006). The combination of the initial and emergent codes yielded the final codebook for our thematic analysis. (The final codebook is found in Appendix 4)

After coding, the final coded transcripts were systematically compared while identifying the most common and recurring concepts, patterns, and themes across the four interviews and mapping them to the Dynamic Capabilities theory. Consequently, in this stage the analysis moved from the description of individual codes and themes by assessing the emerging mechanisms related to the sensing, seizing and transforming capabilities of the four companies (Teece, 2007; Bocken & Geradts, 2020). Overall, the analysis was developed by considering previous literature, coded empirical material, and also new insights that surfaced in all of the interviews (Braun & Clarke, 2006).

3.4.2. Practical documentation

Software: OneDrive was used to store transcripts, codes, and analytic memos.

Audit trail: Versions of the codebook, coding memos, and meeting notes from discussions regarding coding were saved to create auditable decision trails (Nowell et al., 2017).

Quotations: Selected anonymized quotations were used to illustrate themes; identifying details were removed to preserve confidentiality.

3.4.3. Methodological justification

A hybrid deductive-inductive thematic approach was appropriate for this study because it allowed the analysis to stay grounded in existing research while remaining open and sensitive to context-sensitive empirical insights. The coding process was carried out in clear stages and documented, thus strengthening the transparency and rigor of the study (Braun & Clarke, 2006; Fereday & Muir-Cochrane, 2006).

3.5. Trustworthiness and quality of the study

To strengthen the rigor and the quality of this thesis study, the trustworthiness criteria proposed by Lincoln and Guba are used throughout the research (Enworo, 2023). These criteria are credibility, transferability, dependability, and confirmability (Ang et al., 2016). These are commonly used for the quality and rigor assessment of a qualitative study. Therefore, using them was a purposeful choice to ensure that the study meets the standards that qualitative research of this kind is expected to meet (Im & Chee, 2006; Lemon & Hayes, 2020).

3.5.1 Credibility

Credibility refers to the level of confidence that can be placed in the truth of the findings and how accurately they reflect the actual perspectives of the participants (Ang et al., 2016; Lemon & Hayes, 2020). To strengthen the credibility of this study, several measures were put in place.

After identifying the themes and coding the interviews, the final codebook and the coded interview transcripts were sent to the four interview participants so they could confirm

whether the interpretations actually captured their intended meanings. This reduced the risk of misunderstanding or taking statements out of context (Haddad et al., 2020). All of the participants responded with confirmations, justifying the codebooks and the coded transcripts.

The research material and the patterns that were emerging from it were also brought into discussions with colleagues and external reviewers. This was done to identify any potential researcher bias and to reduce the risk that interpretations only reflected the researcher's own assumptions (Behjati, 2017; Kars-Unluoglu & Kevill, 2021).

3.5.2 Transferability

Transferability is about how much the findings of this study can be applied or extended to other contexts, for example other manufacturing sectors or different SME environments (Ang et al., 2016; Enworo, 2023). This qualitative research does not aim for statistical generalizability, transferability can still be supported by providing sufficient contextual detail. In this study, it is addressed through detailed description and purposeful sampling.

Detailed descriptions are provided of the research context, the SMEs that participated in the study, and the specific environmental transitions that were being examined. This level of detail gives readers information to assess whether the findings could be relevant to their own context (Kars-Unluoglu & Kevill, 2021).

The organizations were selected in a way that brings together a range of different perspectives on sustainability barriers (or experience in sustainability in general) and Dynamic Capabilities, rather than only capturing one type of experience or one kind of firm (Kars-Unluoglu & Kevill, 2021).

3.5.3 Dependability

Dependability is about making sure that the data and the findings are stable and consistent over time and across different conditions. It is achieved by keeping the research process transparent and logical so that another researcher could follow them and understand how every decision was made (Im & Chee, 2006; Enworo, 2023). In this study, dependability is supported through audit trail and protocol.

Every step of the research process was documented, including the raw data and all of the coding decisions that were made along the way, so that the full trail of the analysis is visible and traceable (Nowell et al., 2017).

A standardized protocol was used for the interviews and the data collection process to make sure that each SME case was approached in a consistent way and that none of the cases was treated differently from the others (Caldera et al., 2019; Haddad et al., 2020).

3.5.4 Confirmability

Confirmability is about making sure that the findings are neutral and that they reflect the actual experiences of the respondents rather than being shaped by the researcher's own biases or personal interests (Ang et al., 2016; Lemon & Hayes, 2020). In this study, confirmability is established through reflexivity and chain of evidence.

A critical self-reflection was maintained throughout the study, continuously questioning the role of the research team and the potential influence it might have had on the phenomenon being investigated (Behjati, 2017).

A clear and traceable link is provided between the final findings and the original interview data, making sure that key interpretations were connected to what the participants actually lived and experienced (Caldera et al., 2019; Enworo, 2023).

3.6 Ethical considerations

Research ethics play a very important role in conducting high quality research. Ethics are particularly important when interviews are used as a data collection method because personal perspectives and confidential information relating to organizations will be collected during these interviews. (Keith & Fairfield, 2017; McGregor, 2023)

3.6.1 Informed and voluntary participation

Prior to each interview, it was made clear to the participants why this research was being conducted, what topics would be covered during it, and that data could only be used for academic purposes. The reason for doing so was to make sure that the participants were aware of the aim of the research and their participation was not driven by other confusing aspects. The participants provided their informed GDPR consent before taking part in the interviews. (Keith & Fairfield, 2017; European Commission, n.d.)

The participation was strictly voluntary, and the informants were informed about the option not to answer any question, as well as their ability to leave the conversation at any time with no ramifications. This is especially relevant regarding the current study because the issue of sustainability transition can touch on certain aspects related to the internal processes within the organization, its weaknesses, strategic problems, and even commercial secrets. Thus, giving the subjects the opportunity to determine the level of openness was necessary for ethical considerations. (McGregor, 2023)

3.6.2 Anonymity, confidentiality, and data protection

An important ethical principle of this study was to maintain the confidentiality and anonymity of the participants which is essential in a qualitative study (Lim, 2025). A clear communication about confidentiality was established and maintained with the participants which is also an essential aspect (DiCicco-Bloom & Crabtree, 2006).

Therefore, all participants were anonymized as well as their companies. The names of the participants and the companies were removed, as well as every identifying detail from the transcripts and replaced by general names such as Participant 1 and Company 1. Some commercially or strategically sensitive contextual details were also generalized in the transcripts. It was essential because the interviews discuss barriers, operational limitations, and several sensitive aspects in a specific industrial sector. It was also important that the authors anonymized the transcripts before the AI-supported language review to ensure that no sensitive company-related and personal information was provided to the AI tool. Overall, removing these details reduced the risk of experiencing negative consequences for both the companies and participants. (McGregor, 2023; European Commission, n.d.)

Proper handling of interview-related documents was also ensured. Audio records, interviews, and transcriptions were kept safe, and used solely for purposes of this study. This was instrumental in ensuring proper handling of research data and helped in minimizing any chances of information being accessed, disclosed, or misused. Ethics in this study were thus not just part of procedures that had to be followed as a matter of formality, but an important consideration for carrying out ethical, reliable and credible research (Keith & Fairfield, 2017; European Commission, n.d.).

The interview recordings and transcripts were strictly used only for the purposes of this research study. The recording and transcription files were securely stored for the entire research period and access to these files was limited to the researchers only. Files and data were handled in a way which is consistent with the expectation that participants and their companies should always be prevented from unnecessary harm or exposure. The files and data were deleted after the completion of the research study.

3.7 Methodological limitations

First, it should be noted that the research was conducted through a qualitative approach, which implies that its objectives include developing insights and contextualizing results but not making them generalizable in terms of statistics. Thus, the results do not imply that the described mechanisms and trends are applicable to all manufacturing SMEs. Instead, the primary objective is to create a deeper understanding of how the mentioned barriers are perceived and addressed in specific industrial settings.

The second limitation is the low sample size used in the study. Due to the relatively low number of respondents in the research process, there are limitations regarding the empirical basis, which needs to be taken into account. The relatively low number of respondents empirically limited and geographically concentralized this study to Hungarian companies. This geographical limitation may limit the transferability of the findings to other industrial contexts, sectors, or larger steel and metal manufacturing SME populations. However, the assessed SMEs operate within EU-wide supply chains, under European sustainability-related expectations.

A third limitation is the possibility of access bias. Companies that agreed to take part in studies about sustainability might, in some cases, be generally more receptive towards the issue, more introspective, or more willing to think about it, compared with those who decline to get involved in the study. Therefore, the findings of this study may reflect the perspectives of SMEs that are more advanced in sustainability strategies, or more aware of sustainability-related challenges.

In addition, the research draws on perceptions that participants formed about their organizational context based on interviews. This means that the findings are based on what participants perceive, interpret, and communicate regarding their organizational

context, and this does not necessarily align with reality. All of the participants of this study were CEOs, owners, or executive-level representatives, therefore the interview data may be colored by selective memory, subjective interpretation, or the need to portray an organization in a positive light. On the other hand, these perceptions are valuable for the purpose of this study because sustainability transitions are not just the function of factual circumstances but also of how decision-makers interpret them.

Finally, during the research, interviews took place at one point in time rather than following organizations through a period of time making it cross-sectional in nature. Since the transition to sustainability is a complex process and Dynamic Capabilities develop over time, this cross-sectional approach cannot fully capture how the barriers and capabilities change or evolve within the studied SMEs.

4 Empirical Findings

This section presents the empirical findings of the four semi-structured interviews with the four steel and metal manufacturing SMEs. In this section the most relevant findings are presented according to the research question and purpose of this thesis. The empirical findings are presented in a descriptive way, focusing on the participant's expressions and experiences.

4.1 Sustainability Awareness and External Pressures

Each of the studied SMEs were aware of environmental sustainability, however, this awareness was usually triggered externally through customer requirements, energy price shocks or certification and reporting obligations. This suggests that sustainability sensing capabilities were not yet developed as a systematic internal capability in the studied SMEs.

Participant 1 highlighted that there were external pressures to integrate sustainable solutions into Company 1's operations. However, these pressures were not strong enough to force strategic level turns in the organization, therefore, sustainability was not perceived as a major strategic risk or opportunity: *"I do not see that sustainability issues currently represent a major strategic risk for us, but I also do not see them as a major strategic opportunity. In our case these are currently not so strong that they would force a strategic-level turn".*

Participant 3 framed sustainability as an entry condition in the future: *"I would rather say it is more and more present in the background as an expectation or as a future entry condition"*

Participant 4 experiences a lot of pressure from the customers' side as they are requiring EcoVadis certifications and progress in sustainable practices: *"both customers XY and YX well, let's say especially YX is pushing this strongly. They required us to register on a platform run by a company called EcoVadis"*.

Overall, customer requirements appeared as one of the most prominent external factors that are influencing how the studied SMEs become aware of sustainability pressures and opportunities.

4.2 Environmental Data Collection and Measurement

Data collection and measurement appeared to be one of the key internal barriers in the studied companies. Different levels of environmental data collections were described by the participants as some measured certain indicators already and others were only starting to collect sustainability data.

Participant 1 highlighted that Company 1 did not measure its carbon footprint because they did not have a *"separate emission reporting system"*, due to the lack of internal capacity and expertise.

According to Participant 2, Company 2 was actively measuring environmental data, but without a unified system: *"Well, we are measuring but we do not have a uniform practice for it"*.

Participant 3 indicated a similar challenge by explaining that Company 3 collected data on their energy consumption, energy costs, and system parameters, however, they were lacking a regular dashboard system for it: *"we still do not have a mature, regular sustainability dashboard or something like that"*.

Lastly, Participant 4 pointed out that Company 4 was at the beginning of sustainability data collection, causing a delay in their carbon reduction strategy as they were lacking the sufficient capacity and experience to start earlier: *"We haven't developed, because so far we haven't had the capacity for it."* The same participant also added: *"And in practice we are now starting to collect the data"*.

4.3 Financial and Operational Considerations of Sustainability Investments

The interviews reveal that the examined SMEs can actually benefit from certain sustainability opportunities, but only those that align with logical financial and operational considerations. Sustainability efforts occur mostly when the environmental

gains coincide with reductions in costs, increases in efficiency, retention of customers, and receipt of external funding.

Participant 2 directly connected sustainability to financial rationality: *"We want to be sustainable but not at any cost. We are constantly looking for those tools and opportunities where the fact that we are green pays off. In the short or long term, but there is some kind of return on these things."*

Participant 1 also explained this financial filter, adding operational questions as essential decision-making aspects: *"So for us sustainability exists, but inside a very strongly return-oriented and operationally limited framework. In other words, we are not against sustainability. We are trying to act where it is possible. But the possibility is always filtered through the question: can we afford it, and what will happen to the company operation if we do it? That is the real situation."*

Participant 3 linked this financial selectivity to survival pressure: *" simply surviving as a company without going bankrupt, is a much more immediate question right now. It is more about survival."*

Participant 4 also tied sustainability decisions to the financial calculation: *"so if, for example, energy is not cheaper because of it, and only incidentally carbon becomes cheaper and carbon emissions incidentally decrease then we could not take that step, because we simply do not have such financial resources."*

The interviews also suggested that external funding can support sustainable development in some cases, positively influencing sustainability investments. Participant 4 introduced an investment opportunity in solar panels for which Company 4 received state support: *"The decision was basically put into a checkmate position by the fact that there was 50 percent state support available for it. So we really only had to pay half of it, and the support was non-repayable."*

However, Participant 3 described another side of support programs, pointing out that they added uncertainty: *"There are few of them, they are complicated, and very often the whole process is uncertain. There is a huge amount of administration, it takes a lot of time, and in many cases it is not even clear what will finally come out of it."*

Overall, several interview participants described that operational efficiency, cost savings, and production improvements were the main decision-maker factors in sustainability-related actions.

Participant 3 described a strong connection between sustainability investments and the importance of efficiency improvements: *"In the same way, better process stability, lower scrap, longer tool life, or needing fewer later corrections are all things that are at the same time efficiency benefits and sustainability benefits."*

The same logic appeared in Participant 4's explanation: *"if there is no purpose for which we need to invest in a machine, then we do not invest just to reduce our environmental or carbon footprint."*

4.4 Organizational Roles, Responsibilities, and Routines

Limited sustainability formalization within the studied organizations was identified. Several of the studied SMEs explained that they did not have dedicated sustainability departments, officers, clear routines, or established systematic strategies.

Participant 1 described: *"But still, it is important that this is not formalised in our company as a separate strategic system. We do not have a separate sustainability department, we do not have a dedicated sustainability officer, and we do not have a fully developed sustainability roadmap or something like this. So at the moment sustainability is present more as a management-driven intention than as a formal organisation-wide system."*

Participant 4 explained a similar situation: "*at the moment we still do not have a developed strategy, neither environmental nor let's say ESG/environmental. And for now, the long-term strategy affecting the operation of the company still only exists in my head. But it is not broken down into annual plans, and the environmental part has not yet been laid down in a strategy."*

Participant 1 highlighted that no major structural changes were made for sustainability projects or investments: *"So there are no really new roles, there is no separate sustainability officer, and the internal organisational structure has not changed in a significant way because of this. It is more that certain measures have become part of the*

operation, but from this no separate organisational logic or structure was created around them."

Last but not least, Participant 3 connected limited human resources to Company 3's hindered sustainable development: *"At the same time, if I look at sustainability as a separate field energy optimization, carbon footprint calculation, regular ESG data handling, things like this then clearly there are gaps. And the problem is that in an SME few people wear many hats."*

4.5 Production Systems and Operational Constraints

Production systems were described as important aspects when participants introduced sustainability-related changes. They suggested that the changes in production must align with employee skills, quality expectations, delivery reliability, cost control, and existing operational systems.

Participant 3 expressed that it is hard to change production processes without taking risks while new technological solutions cannot be evaluated solely through energy consumption: *"production is not a laboratory game"; "Even then you cannot only look at how much energy they consume. You also have to ask whether our people understand them, whether they can operate them, whether they fit into the existing system, how long the introduction takes, how much downtime there is, and how much risk it brings."*

As Participant 4 explained, production purposes must be considered when they evaluate investments in new machinery: *"if there is no purpose for which we need to invest in a machine, then we do not invest just to reduce our environmental or carbon footprint."*

Overall, the findings indicate that operational stability, machine compatibility, employee knowledge, and financial risks were considered when the studied SMEs made sustainability related production decisions.

4.6 Sustainability Reporting and Certifications

An important theme that appeared across the interviews were related to sustainability reporting and sustainability certifications. Participants explained that customer expectations, certification systems, administrative workload, and regulatory requirements are closely connected to sustainability reporting.

According to Participant 3, occasional environmental measurement is a major challenge for Company 3: *"the problem is that these are occasional measurements, not continuous ones. Not because they are not important, but because we simply do not have a separate person for this, and we do not have the capacity to take it to the depth that maybe would be needed professionally."*

Participant 1 stated directly that reporting tasks are often seen internally as something burdensome for the organization and something that has to be submitted on time, rather than as useful information for long-term development: *"In an SME this can be quite a burden, especially if there is no dedicated person for it."; "In other words, inside the organisation they often appear as, this has to be submitted or this has to be done on time, not as, this is a useful data source for long-term development."*

As Participant 2 suggested that certification systems are resource intensive processes that are not built for SMEs: *"those large international certification systems that are operating, they are fundamentally built on the logic of larger corporations, therefore measuring and reporting by these systems requires a great deal of resources."* The same participant added that the administration demands for these certifications are extremely extensive: *"practically such a monster documentation a process must be built and operated in order to ensure compliance."* Participant 2 also views sustainability reporting as not an actual sustainability process, indicating that Company 2 is skeptical about these requirements: *"this whole methodology is not about you actually being sustainable, but about having a lot of documents with which you pretend to be sustainable."* Despite this skepticism, Participant 2 also feels the positive effects of sustainability reporting: *"this specifically gives a kind of robustness to the whole thing, and after all, at the end of the day, there is a credibility and a foundation to the designation itself thanks to this, and for this reason. And the other half is really that we see that the classification is determined at the end of the day based on comparison to a peer group. So you are forced to continuously improve in order to be able have a good certification"*

Participant 4 explained a distinct reporting system called EcoVadis as customer-driven, indicating that external customer expectations play a key role in Company 4's sustainability reporting: *"reporting on the EcoVadis platform was requested by (partner name)."*

4.7 Customer Requirements and Market Access

As discussed in the previous sections, customer requirements were described as important external pressures in the interviews. Customer relations, supplier status, and future market access are strongly connected with sustainability by the participants.

According to Participant 4, ESG-related activities are closely related with future business opportunities and supplier status: *"On the one hand, that makes it easier to maintain this supplier status, and on the other hand it may also help us gain new work."..."what is still a nice plus today may become a minimum requirement tomorrow."*

Based on the interview with Participant 4, when advancements in ESG services are clearly communicated, they may become advantages when Company 4 deals with companies in the Western European region.

4.8 Summary of Empirical Findings

Overall, the empirical findings indicate that the SMEs studied in this thesis are aware of environmental sustainability opportunities and challenges, however, their responses and processes are different. External factors such as customer requirements, certification and reporting obligations, energy prices and regulations are closely connected to these SMEs' sustainability awareness. It is also revealed by the findings that although data collection exists in the studied SMEs in different forms, they are often fragmented, irregular, or not supported by uniformized systems. Sustainability investments are often influenced by financial considerations, especially in relation to affordability, payback time, external funding, and operational relevance.

Sustainability actions were described in connection with efficiency, energy savings, waste reduction, customer requirements, and production improvements within the assessed companies. The interviewees also emphasized that sustainability was not formalized through dedicated roles, departments, or organizational routines.

Furthermore, production systems were described in these SMEs as difficult to change due to the fact that they might affect quality, delivery, employee knowledge, downtime, cost, and operational stability. Lastly, the interviewees framed sustainability reporting and certifications as increasingly important but also administratively demanding.

5 Analysis

This section analyzes the empirical findings. It explains and synthesizes the identified internal, external, and interface barriers while examining how they constrain the sensing, seizing, and transforming capabilities of the studied SMEs. The analysis is also connected to the frame of reference. Lastly, the section also presents an empirical framework which describes the main relationships between the identified mechanisms.

5.1 Fragmented Sensing of Sustainability Pressures

The findings suggest that all four studied SMEs were aware of environmental sustainability, however, this awareness was usually triggered externally through customer requirements, energy price shocks or certification and reporting obligations. Consequently, these companies did not develop their sustainability sense as a systematic internal capability. This aligns with prior literature indicating that SMEs face increasing external pressures from customers, regulatory bodies, and supply chains (Alayón et al., 2022; Hąbek, 2025; Neri et al., 2021). On the other hand, the findings of this study also show that these external pressures do not always create strong sensing capabilities in the studied SMEs.

There were several internal barriers that constrained the studied SMEs' sensing capabilities, including the lack of limited expertise, the lack of dedicated responsibilities, and the use of weak data collection and monitoring systems. These findings on internal barriers align with the findings of prior studies (Hansen et al., 2025; Müller et al., 2024; Trianni et al., 2019). Consequently, the empirical findings are consistent with the frame of reference. However, from the perspectives of sensing, these findings also suggest that the lack of these systems and resources constrains the assessed companies' ability to understand their environmental impacts and recognize sustainability-related opportunities.

Furthermore, the findings also add nuance to prior research by suggesting that sensing is also shaped by where the assessed SMEs are positioned within their supply chains. In cases where a downstream customer demands that the company operates with a high level

of environmental sustainability, presenting certifications and reporting on their progress, it becomes visible as a strategic opportunity and pressure. On the other hand, when the customers' requirements are lower regarding these sustainability aspects, sustainability remains secondary. Consequently, internal operations, resources and external pressures not only act as barriers in the assessed SMEs' sustainable development processes. They are also limiting factors in developing sustainability-related sensing capabilities as they play a key role in whether the assessed companies can identify and interpret these opportunities or threats, making sensing fragmented among the assessed companies.

5.2 Selective Seizing Under Financial Pressures

The interviews reveal that the examined SMEs can actually benefit from certain sustainability opportunities, but only those that align with logical financial and operational considerations. Sustainability efforts are thus selective. They occur mostly when the environmental gains coincide with reductions in costs, increases in efficiency, retention of customers, and receipt of external funding.

Therefore, these findings align with previous studies that indicate limited financial resources, high upfront investment costs, uncertain returns, and short-term survival pressure as hindering factors in SMEs sustainable development (Alayón et al., 2022; Neri et al., 2021; Sohns et al., 2023). This pattern is reflected in the findings. Environmental value is not the only evaluation aspect when it comes to sustainability opportunities in the assessed companies. Risks, operational relevance, payback time, and affordability are also considered when these companies filter through these questions. Other prior findings arguing that customer demand on sustainability can be price-sensitive (Muruganantham et al., 2025; Qureshi et al., 2023) are also supported by the findings of this study.

However, the findings also extend prior literature, suggesting that the existence of internal barriers in the context of seizing is not solely related to the availability of funds among the studied SMEs. Rather, it is linked to their operational logic. Since there are limitations regarding funding and there is a necessity of maintaining operational continuity, investments towards sustainable development compete against investments toward the core business. Where no clear return was visible on sustainable investments, they tended to be put on hold until further notice. This shows that in the studied SMEs' case, efficiency, cost control and survival still have priority over long term sustainability

innovation. Therefore, these companies' seizing capabilities are selective, meaning that they are not focusing on opportunities that do not offer financial profitability, operational stability, customer retention, or external support.

5.3 Partial Transforming in Rigid Production Systems

The interviews and the findings indicate that transformation appeared to be the most limited Dynamic Capability among the SMEs analyzed. They introduced several practices related to sustainability, that are mostly connected to energy efficiency and reporting, but these are not part of their formal strategies, not assigned to specific people in the company, and overall, not part of their organizational routines.

This is connected to the literature arguing that SMEs should not be treated as "little big businesses" (Alayón et al., 2022). The studied SMEs operate without dedicated sustainability departments and sustainability officers while lacking specialized internal expertise. This is in alignment with studies that argue that SMEs are structurally different from larger organizations, usually operating with limited resources, flat organizational structures, and focusing on day-to-day survival (Alayón et al., 2022; Mengistu & Panizzolo, 2023; Sohns et al., 2023).

The findings of this empirical study are also consistent with prior literature that highlight the specific challenges of the steel and metal manufacturing sector. These studies indicate that these companies are constrained by resource-intensive processes, expensive machinery, and operational dependence on stable production systems (Daroń & Górska, 2019; Hąbek, 2025; Neri et al., 2021). This is reflected by the empirical findings which show that environmental benefits are not the only evaluation factors in production changes. Machine compatibility, employee skills, product quality, delivery reliability, and operational risk factors are all considered when deciding to transform the companies' operations assessed in this empirical study.

The empirical findings also added to existing research as they show that transformation in the case of SMEs can remain partial even when sensing and seizing are present. The interviewed companies are aware of the sustainability pressures and opportunities, and they also selectively implement some improvements, however, this does not automatically mean that sustainability becomes embedded in the firms' routines,

structures or long-term strategic processes. Analytically, this means that sustainability transformation should be treated differently from sustainability actions.

5.4 Interface Capability Bottleneck: Sustainability Reporting

The frame of reference identifies sustainability reporting as an interface between internal and external barriers. It is usually the external actors (regulators, customers, banks, etc.) who set the requirements for reporting. Yet to fulfill these expectations, SMEs need well-developed internal data systems, knowledge, expertise, capacity, and strong coordination (Hąbek, 2025; Martinčević et al., 2025; Rantala et al., 2025). This means that reporting becomes one of the clearest areas where external expectations and internal capability gaps meet (Alayón et al., 2022; Hansen et al., 2025; Zhu et al., 2025).

The findings of this study support this interpretation as reporting appeared as a key area where external pressures and expectations meet internal limitations. The studied SMEs indicated increasing requirements to measure and provide environmental data, comply with regulations, and acquire sustainability certifications. However, these companies often lack the needed internal resources to succeed.

Based on the findings, reporting affects all three Dynamic Capabilities. First, it affects sensing, because firms have to identify, collect, and understand environmental data. Second, it affects seizing, because reporting can be connected to supplier qualification, customer retention, or access to finance. Third, it affects transforming, because reliable reporting requires routines, clear roles, and system integration. (Teece, 2007; Bocken & Geradts, 2020)

The analysis of the empirical findings adds a nuance to prior literature by showing that in the studied SMEs, reporting is not just a limiting but also a supporting factor in capability development. Reporting pushes these companies to collect data, document their processes, and define sustainability targets, but it also acts as an administrative burden which consumes resources, taking them away from other operational improvements.

Consequently, this can be interpreted as a reporting paradox, making sustainability more visible for the studied companies, but it may also shift their attention away from real environmental improvements, making sustainability a compliance activity rather than actual progress. This paradox indicates that sustainability reporting is not perceived as

just an activity for pure transparency and accountability by the studied SMEs. They also perceive these processes as inefficient processes that are designed for larger corporations.

Overall, this analysis suggests that reporting may function as a capability bottleneck for the studied SMEs. It increases the need for sensing, seizing, and transforming, while at the same time showing the limited ability of SMEs to do these things systematically. In this sense, reporting is not only an external requirement. It is an interface mechanism where internal weaknesses and external pressures become connected.

5.5 Cross Capability Synthesis: Barriers to Capability Constraints

The empirical findings suggest that barriers do not only affect sensing, seizing, and transforming capabilities, but also the movement between these steps. This is important as the Dynamic Capabilities theory assumes a process in which firms first notice changes (sensing), then decide how to respond (seizing), and finally adjust their routines and resources (transforming) (Teece, 2007; Bocken & Geradts, 2020). In the SMEs analyzed, this process is often weakened or interrupted.

First, weak sensing reduces the effectiveness of seizing. When environmental data is fragmented, irregular, or mostly collected for external reporting in the studied firms, they often struggle to identify clearly where sustainability improvements are most needed or most valuable. Because of this, investment decisions become more uncertain. For example, when carbon footprint data is missing or only measured occasionally, these firms cannot clearly assess the environmental or financial benefits of possible improvements. This shows that weak internal data systems do not only limit awareness but also make it harder to justify acting.

Second, constrained seizing limits transformation. Even when the studied SMEs see sustainability opportunities, they usually act only if the investment has a clear payback or is important for customers. Because of this, the observed actions in the case of these companies focus on energy efficiency and cost reduction, while other environmental improvements are less developed. Participant 4's case shows this clearly: solar panels and lighting modernization were easier to justify because environmental and economic benefits overlapped, while a storage investment linked to the solar system could not be implemented due to high cost and uncertain payback.

Third, weak transformation feeds back into sensing. When sustainability is not formalized into roles, routines, or systems, data collection remains fragmented and dependent on individual attention. This limits future sensing and makes sustainability work more occasional than systematic. Participant 1 explained that without clear systems, time, responsibilities, and organizational attention, data collection becomes fragmented. This shows that the lack of transformation also weakens the conditions needed for effective sensing in the future.

Fourth, reporting acts as both a source of pressure and a bottleneck across the whole capability chain. It pushes firms to improve sensing, because they need data. It also pressures them to act, as customers and certification systems may require responses. In addition, it creates a need for transformation, since credible reporting requires routines and clear responsibilities. However, when reporting is experienced mainly as paperwork or compliance, it can consume limited resources without leading to real strategic learning. This shows that reporting, as an interface barrier, plays a central role in the overall findings.

The main pattern from the findings can be explained like this: in the assessed cases, external pressure for sustainability appeared to increase faster than the SMEs abilities to develop internal capacities and capabilities needed to respond systematically. Consequently, the studied companies tended to respond with selective actions that are financially reasonable or pushed by external actors, instead of making more systematic changes in the organization.

5.6 Empirical Framework: Barrier-Constrained Sustainability-Related Dynamic Capabilities

This thesis offers a barrier-constrained sustainability-related Dynamic Capabilities framework in Figure 2. It summarizes the analytical pattern that was identified throughout the previous sections. The figure shows how internal and external barriers combine into a dynamic capability-constraining barrier system, limiting the studied SMEs' ability to systematically respond to environmental sustainability challenges and pressures. The framework presents sensing, seizing, and transforming as a connected capability chain instead of treating them as independent mechanisms and outcomes. This capability chain displays how fragmented sensing limits the information base needed for appropriate

decision making, how selective seizing narrows the range of actions that can translate into implementation, and how partial transforming prevents sustainability from being embedded into systems, routines, and organizational structures. Consequently, this framework reflects the findings and analysis sections by visually explaining that environmental sustainability in the studied SMEs is not only constrained by individual barriers, but also by the way these barriers constrain the development of sustainability-related Dynamic Capabilities and the movement between them.

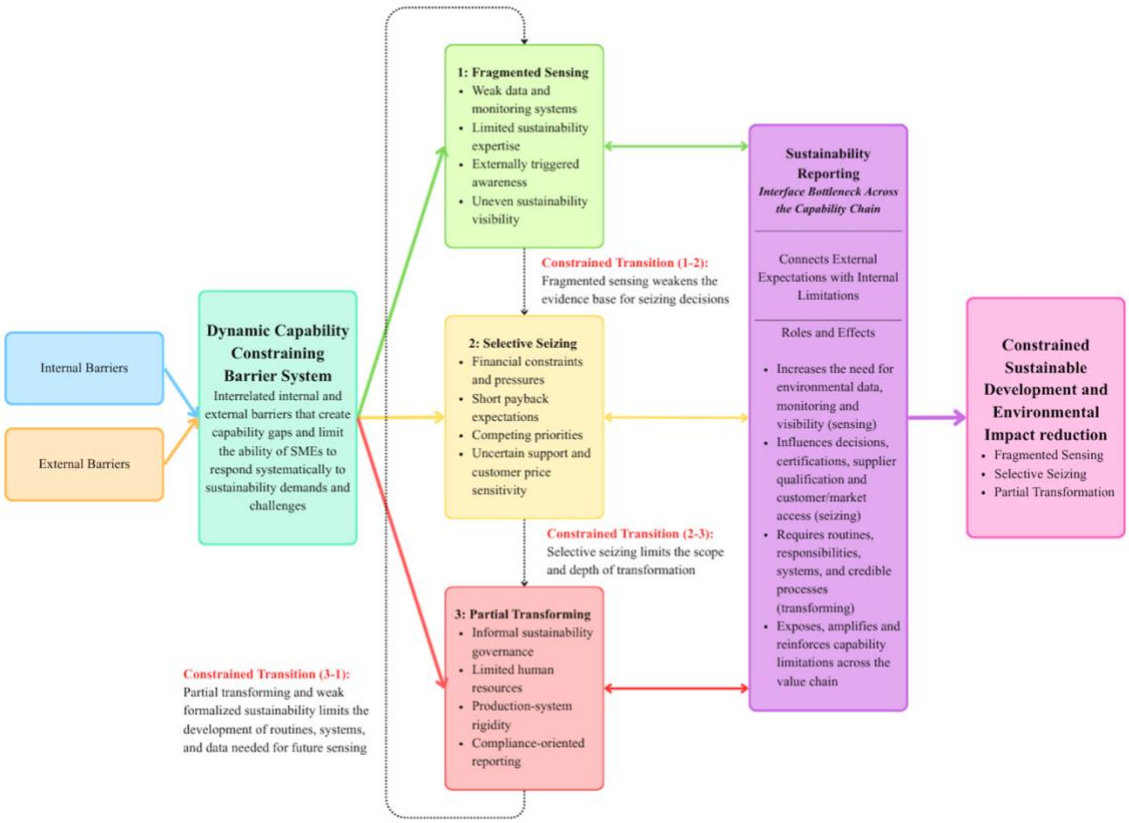


Figure 2 DCCBS Framework Based on the Empirical Findings and Analysis

Different types of relationships are explained with different types of arrows in the framework. The blue and orange arrows from internal and external barriers indicate that they feed into a broader Dynamic Capability Constraining Barrier System. The green, yellow and red arrows pointing from the barrier system to sensing, seizing, and transforming indicate that all of the three capabilities are affected by this system, leading to fragmented sensing, selective seizing, and partial transforming in the assessed SMEs. The black dotted arrows between sensing, seizing, and transforming indicate that the transitions between the different capabilities are constrained. Fragmented sensing

weakens the evidence base for seizing decisions and selective seizing limits the depth and scope of the organizational transformation. The third black dotted line indicates a feedback arrow, emphasizing how partial transforming affects future sensing capabilities as limited routines, systems, responsibilities, and data structures lead to fragmented sensing again. Consequently, the framework indicates a reinforcing cycle rather than a simple one-way process.

The analysis revealed that sustainability reporting appeared to connect external expectations with internal capability gaps in the studied SMEs, therefore it is positioned as a cross-cutting interface bottleneck on the right side of the framework. The findings and analysis also suggest that sustainability reporting both affects and is affected by the Dynamic Capabilities of the studied SMEs. Consequently, the two-way arrows between the Dynamic Capability areas and sustainability reporting indicates this relationship. Lastly, the purple arrow from sustainability reporting toward constrained sustainable development and environmental impact reduction shows the overall outcome of the empirical analysis. Due to the fact that sensing appeared to be fragmented, seizing was selective, and transforming was partial, the studied SMEs' environmental improvement tended to remain limited, uneven, and strongly shaped by the barrier-constrained capability development.

6 Discussion

This section discusses the theoretical contributions and practical implications of the study's empirical findings. It also discusses the limitations of the study while calling for further research in the same research topic.

6.1 Theoretical Contributions

Prior literature identified barriers to sustainable development in manufacturing and steel and metal manufacturing SMEs based on their internal or external origin (Alayón et al., 2022; Neri et al., 2021, Hąbek, 2025). Our synthesis in the frame of reference section revealed that these internal and external barriers are interconnected and reinforce each other. Using the Dynamic Capabilities framework, our empirical study explains that these barriers not just connect to each other, but they also affect the studied SMEs' sustainability-related sensing, seizing, and transforming capabilities (Teece, 2007; Bocken & Geradts, 2020).

In the assessed SMEs, sensing is often limited because sustainability issues are not always identified internally. In several cases sustainability issues were noticed, but usually only after external actors, such as consumers, governments, banks and other stakeholders, raised them. However, the studied SMEs often have limited operations in monitoring their own environmental performance and they appear not to have full understanding of their environmental sustainability related challenges due to missing sufficient skills, data systems, and processes.

The empirical findings and analysis suggest selective seizing among the assessed SMEs since many of the sustainability efforts were made once there was a fit in the company's economic or operational practices. This does not necessarily mean that the issue is not one of will but more likely that decisions in SME companies are influenced by limited resources, the current business focus in the long run and the lack of clear returns from investing in sustainability.

Also, the evidence suggests that transforming appeared to be the most difficult capability to develop for the studied SMEs. The companies that participated in the study have

introduced a range of practices concerning sustainability, but these initiatives were narrowly defined and were not embedded into the strategy, work routines and daily procedures, their responsibilities and their production processes. This connects back to prior arguments that SMEs cannot be simply considered as scaled-down large enterprises because their resources, systems and the context they are acting within is very different from large enterprises (Alayón et al., 2022).

Lastly, one of the main contributions of this thesis is the identification of sustainability reporting as an interface capability bottleneck. While the frame of reference section indicated that sustainability reporting links external demands to internal constraints (Alayón et al., 2022; Hąbek, 2025), our empirical study suggests that sustainability reporting also affects and is affected by the sensing, seizing, and transforming capabilities of the studied SMEs. Therefore, sustainability reporting is more than just the point in which the external expectations of regulators, customers, certification systems, and supply chain partners meet as it can both constrain and foster the development of sustainability-related Dynamic Capabilities in the studied SMEs. Therefore, reporting can encourage SMEs to collect environmental data, clarify and assign responsibilities, and become aware of their environmental impact, however, it may also turn into an administrative burden when firms do not have the necessary capacity, systems, or routines to manage it effectively.

6.2 Practical Implications

From the perspective of SMEs, the results suggest that for sustainability to be relevant, it needs to be associated with operational issues. For the enterprises analyzed in the study, sustainability initiatives were related to energy conservation, material conservation, waste minimization, customer demands, and productivity improvements. From this, it can be seen that sustainability is not distinct from business operations but integral to them.

The results have also indicated that sustainability reporting has taken on more significance among SMEs within extended value chains. Reporting obligations can, however, create problems when companies do not have the appropriate IT infrastructure, expertise, or staff members to answer such obligations. This implies that sustainability reporting should not merely be seen as a regulatory exercise but also one highlighting the limitations of SMEs.

For policy makers and supporting agencies, the results suggest the importance of considering the particular circumstances of SMEs in creating sustainability-oriented legislation and funding mechanisms. While obligations that could be easily met by big companies might prove to be problematic for SMEs because of their lower administrative and financial resources. Thus, the significance of the research lies in its demonstration of the fact that the external demands concerning sustainability should correspond to the internal potential of SME manufacturers.

In the case of larger clients and supply chain players, the implications suggest that the pressures that these SMEs experience in terms of their sustainability could affect the adoption of sustainability practices by such businesses. The client's demand creates pressure for changes, but at the same time, increases the level of administration if such demands are not clear and hard to achieve. Therefore, the transition process towards sustainability among these SMEs involves more than just the firms themselves.

6.3 Limitations

This study has several limitations. Firstly, data was gathered from four Hungarian SMEs operating within the steel and metal manufacturing sector. Although they are operating within EU-wide supply chains, these results may not be statistically generalizable across a larger population of steel and metal manufacturing SMEs in the EU.

Secondly, the data were gathered by interviewing representatives of these companies. This is advantageous for understanding how the obstacles to sustainability are experienced and perceived by companies but does not yield the measures of environmental performance directly from company data.

Thirdly, it is also a limitation that each of the participants in this study were CEOs and owners. Although this provided valuable strategic and executive-level insights, the study does not necessarily reflect the daily operational experiences of the employees involved in different departments where the actual production, measurement, administration, or reporting activities take place.

Additionally, the research is cross-sectional as data were gathered at a single point in time and not longitudinally. Dynamic Capabilities do evolve, and future longitudinal studies

of this nature may further reveal the extent to which sustainability related capabilities develop in SMEs.

Finally, the thesis is limited to environmental sustainability and focuses mainly on barriers. Broader social and economic sustainability dimensions, as well as possible factors that enable sustainability transitions, were not part of the main scope of this study.

6.4 Further Research

Considering the limitations of this thesis, future research might focus on a larger number of SMEs in steel and metal manufacturing across various EU countries, examining the effect of context on sustainability barriers within such an environment. Another type of research that could be undertaken involves longitudinal research to determine how each capability evolves over time.

The other area that could be explored is sustainability reporting, as it was recognized as one of the key interfaces between external pressure and internal constraints. Future research can thus focus on examining whether the adoption of sustainability reporting leads to real environmental benefits or simply increases administrative tasks.

Addressing the participant-related limitation of this study, future research could include participants from different departments of the assessed companies such as production managers, administrative employees, or environmental coordinators to collect and compare daily operational experiences and strategic managerial perceptions.

Lastly, future research may examine the contribution of enablers of sustainability in SMEs, which involve customer involvement, government funding, technology, management commitment, and workforce participation.

7 Conclusion

The purpose of the thesis was to explore the sector specific barriers constraining the environmental sustainability of EU steel and metal manufacturing SMEs while considering the Dynamic Capabilities theory. Hence, the research question addressed was:

How do barriers constrain the environmental sustainability of EU steel and metal manufacturing SMEs in relation to sensing, seizing, and transforming capabilities?

The study concludes that the assessed companies' environmental sustainability is constrained by a complex, interconnected barrier system, limiting development and movement between their sensing, seizing, and transforming capabilities. The study revealed that from the perspectives of Dynamic Capabilities theory, the sustainability-related barriers may cause fragmented sensing, selective seizing, and partial transforming in the studied companies. Fragmented sensing comes from the fact that sustainability awareness is often triggered externally in these firms while having limited internal data collection systems and routines. The studied SMEs' seizing is constrained by competing interests as they often take sustainability-related actions only when they offer clear financial or operational benefits. Lastly, sustainability improvements are not entirely embedded in strategies, production systems, and routines, leading to partial transformation.

The study also suggests that sustainability reporting acts as an interface bottleneck. It bridges EU-wide sustainability-related expectations with internal capacity limitations in the studied companies. Although it may increase sustainability awareness and pressure for action, it can become a compliance burden when internal systems and resources are insufficient.

Answering the research question, the sustainable development of the studied SMEs is not only hindered by the identified barriers. It is hindered by the way these barriers constrain the development of sustainability-related Dynamic Capabilities and weaken the capability chain from sensing to seizing and to transforming.

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Appendix

Appendix 1 - Interview Guide



JÖNKÖPING UNIVERSITY
Jönköping International Business School

Interview Guide

I. Backgrund:

1. Could you please introduce your company briefly? (size, main activities, markets)
2. What is your position within the company and what responsibilities do you have?
3. How would you describe the company's current overall approach to sustainability?
4. When did sustainability become relevant for the company, and what initiated this process?

II. Current Sustainability Practices

1. What specific sustainability-related measures have been implemented so far? (e.g. energy efficiency, waste reduction).
2. Do you currently measure energy consumprion, waste volumes, or emissions?
    a. If yes, what methods do you use?
3. Have you experinced any tangible benefits from sustainability efforts? (e.g. cost savings, new customers, competitive advantages, increased efficiency).

III. Constraints and Challenges

1. Is there a competition between sustainability-related investments and other urgent business priorities within your company?
2. Did you experience any disruprion in production schedules or daily operations due to the integration and testing os new energy efficient solutions?
3. What is/are the biggest barrier(s) or challenge(s) that makes the adoprion of environmentally sustainable technologies difficult?
4. Is your company subject to any sustainability-related reporting requirements or administrative obligations?

IV. Internal Capabilities

1. Do you feel that the company and its employees have sufficient knowledge to successfully adopt more advanced sustainability practices?
2. Do you face any challenges in collecting and managing environmental data?
3. Do you feel that your company has sufficient technological infrasturcture to adequetly support the sustainability transition?

V. Sensing – Perceiving External Environment

1. How does the company sense new sustainability-related regulations, market expectations, or technological opportunities?
2. To what extent do you feel that environmental or sustainability issues represent a strategic risk or opportunity for the company?
3. Are ou aware of any external events that triggered or accelerated any sustainbility initiative?

VI. Seizing – Capturing Opportunities

1. How do you decide whether to implement a sustainability investment, especially if it involves short-term cost increases?
2. What role does top management play in making sustainability-related decisions?
3. Have there been cases where a sustainability opportunity could not be pursued due to lack of resources or uncertainty?

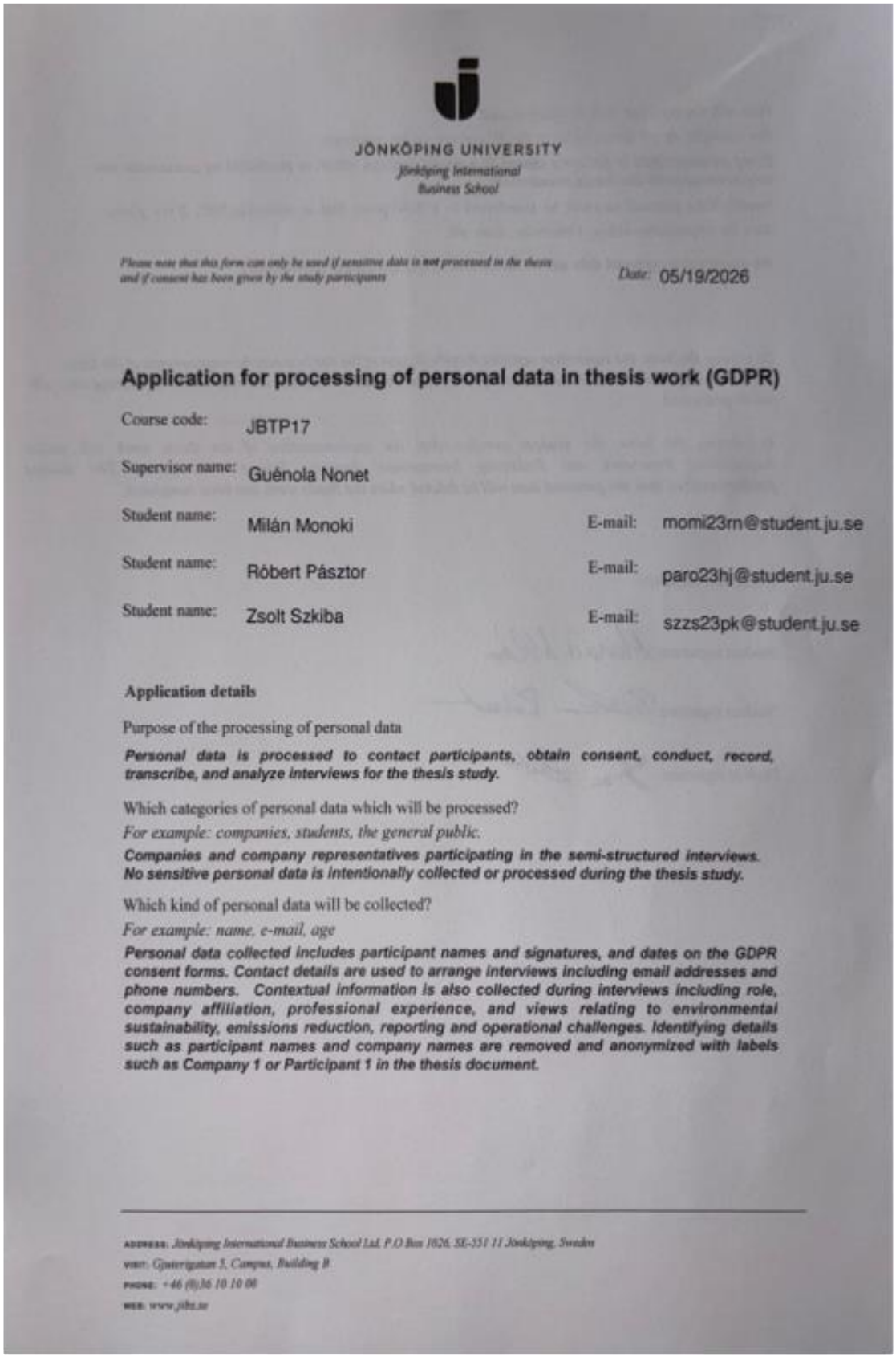
VII. Transforming – Organizational Transformation

1. Have internal processes, roles, or responsibilities changed as a result of sustainability efforts?
2. To what extent do you feel the company is capable of sustaining and further developing its current sustainability practices in the long term?
3. What internal capabilities (e.g. knowledge, data management, technology) are most lacking for further progress?

VIII. Closing Question

- If you had to highlight one single factor that would most help the company in its sustainability transition, what would it be?

Appendix 2 - Application for Processing of Personal Data in Thesis Work



JÖNKÖPING UNIVERSITY
Jönköping International Business School

Please note that this form can only be used if sensitive data is ***not*** *processed in the thesis and if consent has been given by the study participants*

Date: 05/19/2026

Application for processing of personal data in thesis work (GDPR)

Course code: JBTP17

Supervisor name: Guénola Nonet

Student name: Milán Monoki E-mail: momi23rn@student.ju.se

Student name: Róbert Pásztor E-mail: paro23hj@student.ju.se

Student name: Zsolt Szkiba E-mail: szzs23pk@student.ju.se

Application details

Purpose of the processing of personal data

Personal data is processed to contact participants, obtain consent, conduct, record, transcribe, and analyze interviews for the thesis study.

Which categories of personal data which will be processed?

For example: companies, students, the general public.

Companies and company representatives participating in the semi-structured interviews. No sensitive personal data is intentionally collected or processed during the thesis study.

Which kind of personal data will be collected?

For example: name, e-mail, age

Personal data collected includes participant names and signatures, and dates on the GDPR consent forms. Contact details are used to arrange interviews including email addresses and phone numbers. Contextual information is also collected during interviews including role, company affiliation, professional experience, and views relating to environmental sustainability, emissions reduction, reporting and operational challenges. Identifying details such as participant names and company names are removed and anonymized with labels such as Company 1 or Participant 1 in the thesis document.

ADDRESS: *Jönköping International Business School Ltd, P.O Box 1026, SE-551 11 Jönköping, Sweden*
VISIT: *Gjuterigatan 5, Campus, Building B*
PHONE: *+46 (0)36 10 10 00*
WEB: *www.jibs.se*

How will the personal data be safely stored?

For example, in my home folder at the JU servers, on my computer.

Every personal data is securely stored in a digital storage which is protected by passwords and only accessible for the thesis researchers

Specify if the personal data will be transferred to a third party, that is, otherthan JIBS. *If yes, please state the organizationbelow. Otherwise, state no.*

No identifyable personal data will be forwarded to any third party outside JIBS

By signing the form, ***the supervisor*** *certifies that the design of the thesis meets the requirements of the Data Protection Regulation. This includes that informed consent will be obtained and that sensitive personal data will not be processed.*

By signing the form, ***the student*** *certifies that the implementation of the thesis work will follow the approved framework and Jönköping International Business School's instructions.* ***The student further certifies that the personal data will be deleted when the thesis work has been completed.***

Supervisor signature

Student signature

Student signature

Student signature

Appendix 3 - Initial codebook for Interview Transcripts

Aggregates	Themes	Concepts	Code Number
Internal Barriers	Resource and Operational Constraints	Financial constraints	IB1
		Limited human resources	IB2
		Lack of sustainability knowledge/expertise	IB3
	Operational and Structural Constraints	Weak data systems / digitalization gaps	IB4
		Fragmented processes	IB5
		Legacy equipment / technological rigidity	IB6
	Leadership Focus	Management hesitation / resistance to change	IB7
		Short-term operational focus	IB8
External Barriers	Institutional and Market Pressures	Regulatory complexity / uncertainty	EB1
		Customer pressure and price sensitivity	EB2
		Market uncertainty	EB3
	Structural and Resource Constraints	Limited access to external finance/support	EB4
		Infrastructure limitations	EB5
		Supply chain dependence	EB6
Interface of Internal and External Barriers	Reporting Constraints	Sustainability reporting pressure	RI1
		Reporting capability gaps	RI2
		External data dependency	RI3
Sensing	Environmental and Regulatory Sensemaking	Limited awareness of sustainability risks/opportunities	SEN1
		Difficulty interpreting regulations	SEN2
	Weak Information and Monitoring Systems	Weak environmental monitoring	SEN3
		Weak scanning of market/technology developments	SEN4
Seizing	Financial Risk and Investment Uncertainty	Hesitation to invest	SEI1
		Uncertain ROI	SEI2
	Strategic Decision Constraints	Competing priorities	SEI3
		Risk aversion in decision-making	SEI4
Transforming	Operational Rigidity	Difficulty changing production systems	TR1
	Limited Integration and Organizational Inactivity	Organizational inactivity	TR2
		Weak integration of sustainability into operations	TR3
	System-level Constraints	Supply-chain constraints on systemic change	TR4
		Need for structural reconfiguration	TR5

Appendix 4 - Final Codebook for Interview Transcripts

Aggregates	Themes	Concepts	Code Number
Internal Barriers	Resource and Operational Constraints	Financial constraints	IB1
		Limited human resources	IB2
		Lack of sustainability knowledge/expertise	IB3
	Operational and Structural Constraints	Weak data systems / digitalization gaps	IB4
		Fragmented processes	IB5
		Legacy equipment / technological rigidity	IB6
	Leadership Focus	Management hesitation / resistance to change	IB7
		Short-term operational focus	IB8
External Barriers	Institutional and Market Pressures	Regulatory complexity / uncertainty	EB1
		Customer pressure and price sensitivity	EB2
		Market uncertainty	EB3
	Structural and Resource Constraints	Limited access to external finance/support	EB4
		Infrastructure limitations	EB5
		Supply chain dependence	EB6
Interface of Internal and External Barriers	Reporting Constraints	Sustainability reporting pressure	RI1
		Reporting capability gaps	RI2
		External data dependency	RI3
Sensing	Environmental and Regulatory Sensemaking	Limited awareness of sustainability risks/opportunities	SEN1
		Difficulty interpreting regulations	SEN2
	Weak Information and Monitoring Systems	Weak environmental monitoring	SEN3
		Weak scanning of market/technology developments	SEN4
Seizing	Financial Risk and Investment Uncertainty	Hesitation to invest	SEI1
		Uncertain ROI	SEI2
	Strategic Decision Constraints	Competing priorities	SEI3
		Risk aversion in decision-making	SEI4
Transforming	Operational Rigidity	Difficulty changing production systems	TR1
	Limited Integration and Organizational Inactivity	Organizational inactivity	TR2
		Weak integration of sustainability into operations	TR3
	System-level Constraints	Supply-chain constraints on systemic change	TR4
		Need for structural reconfiguration	TR5
Emergent Results			
Aggregates	**Themes**	**Concepts**	**Code Number**
Sustainability as a Response	Energy and Cost-driven Sustainability	Energy autonomy and renewable energy as the main direction	EM1
		Environmental–economic co-benefit logic	EM2
		Energy-price shock as sustainability accelerator	EM3
	Informal Sustainability Governance	Owner-manager centrality	EM4
		Embedded but non-formalized sustainability	EM5
	Sustainability as External Legitimacy and Market Access	Administrative visibility over practical sustainability	EM6
		Supplier qualification	EM7

Appendix 5 - AI Declaration

DECLARATION

We hereby declare that this thesis represents our own work.

We declare that we have obtained the necessary ethical approval (e.g., GDPR) and acknowledged our obligations and the rights of the participants in the research.

We declare that if we have made use of non-sensitive personal data we have collected *GDPR Thesis consent forms*, including contact details with participants in our study. These forms have been shared with our tutor.

We have read and applied the Jönköping University's current research ethics guidelines with regards to the use of artificial intelligence (AI) tool in this work as presented in the Course Guide. During the preparation of this work the authors used the following AI tools and the purpose for their use.

- We used AI to ensure our thesis is properly structured; after writing a section, we used different AI models (ChatGPT, CoPilot, Claude, Gemini) to analyze it and compare it to the structure in the study guide.
- To ensure the thesis uses a consistent, academically appropriate language, we used Grammarly AI.
- As mentioned in the methodology section, the interview transcripts were also reviewed with ChatGPT only to detect grammar and spelling errors. Before this AI-assisted review, participants and companies were anonymized and commercially sensitive details were manually removed.

As authors, we have reviewed and edited the content as needed and take full responsibility for the content of the thesis.

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